Localized Matched-Subspace Detection for Functional Ultrasound and Ultrafast Doppler Imaging

Arthur Shune*
Independent Researcher

Abstract

Functional ultrasound and ultrafast Doppler pipelines remain vulnerable to nuisance-driven false positives in the extreme score tail, even after conventional clutter suppression. Those rare false positives can distort vascular maps and reduce confidence precisely in the operating region where readers most care about trustworthiness.

We propose a post-beamformed, tile-local detector-head framework for fUS slow-time data. Its simplest and main form uses band-limited matched-subspace scoring after ordinary residualization. A prespecified adaptive variant keeps that score by default and promotes clutter-heavy pixels onto a whitened branch only when residual guard-band energy indicates localized structured nuisance, while a fully whitened covariance-adaptive variant is retained as a specialist mode for regimes in which full covariance adaptation is preferable. The historical connection to STAP is architectural rather than literal: adaptation remains in beamformed slow time, not in array-channel space–time. We additionally study an optional shrink-only regularizer, but the main efficacy claims in this paper are carried by the detector heads themselves. Conditional execution is evaluated as a separate decision-rule variant rather than as a free acceleration.

Quantitative detection claims are restricted to datasets with independent labels or bounded structural references that are fixed independently of the short-window score maps being evaluated: labeled k-Wave brain simulations, a structurally labeled flow phantom, a published localization-derived structural reference on real in-vivo rat-brain IQ, and a prespecified held-out SIMUS/PyMUST moving-scatterer benchmark. On the labeled brain stress tests and the structurally labeled phantom, this detector-head framework retains nonzero sensitivity at operating points where several power-Doppler-style baselines collapse. On the main held-out structural benchmark, the strongest fixed head is the simpler band-limited matched-subspace score on the prespecified residual families. A held-out stress confirmation on the same SIMUS/PyMUST mobile scene family then shows that under cardiac-like pulsation and short-ensemble pressure, the adaptive and fully whitened extensions add value beyond that default on the rows where nuisance pressure is highest. A separate fixed-calibration transfer audit on that benchmark then shows that thresholds learned on separate calibration seeds carry onto held-out seeds without erasing the fixed head's nuisance-control advantage. On real in-vivo rat-brain IQ from ULM Zenodo 7883227, a one-time registration of the published PALA microbubble-localization vascular map back onto the IQ grid yields a bounded structural-label audit in which a prespecified no-registration fully whitened matched-subspace specialist shows a modest but consistent structural advantage over PD and Kasai on a vessel-core versus perivascular-shell endpoint while remaining below the 64-frame acquisition budget on an RTX 4080 SUPER; a complementary fixed-calibration block split retains a small but nonzero held-out core-detection advantage at

*Correspondence: `arthur.shune@gmail.com`. ORCID: `0009-0004-6433-3856`. Work performed in an independent research capacity.

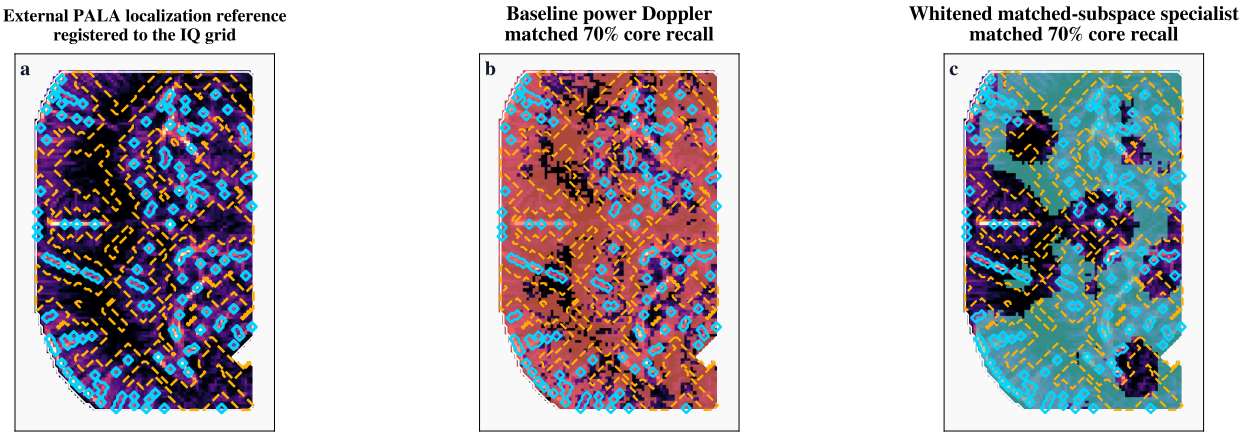

Figure 1: Localization-derived structural-label result on ULM Zenodo 7883227 using the published PALA rat-brain localization rendering as an external vascular reference. Left: the PALA-derived vessel-core and perivascular-shell masks after registration back to the IQ grid. Middle and right: the same audited 64-frame MC–SVD residual window (block 001, frames 128:192), thresholded at matched 70% vessel-core recall for baseline power Doppler and the prespecified no-registration whitened matched-subspace specialist. This figure is intentionally visual; the formal held-out quantitative comparison is reported later in Table 13.

near-target shell FPR. This in-vivo result is about structural vascular recovery, not functional activation or hemodynamic-response estimation.

Overall, the results support a localized matched-subspace detector-head framework for post-beamformed fUS: a fixed default detector with a practical adaptive extension and a bounded covariance-adaptive specialist mode, complementing rather than replacing upstream clutter suppression.

1 Introduction

Functional ultrasound (fUS) is now used for vascular and functional mapping in settings ranging from preclinical neuroscience to awake surgery and cranial-window human imaging. In these settings, the practical problem is often not average sensitivity alone. A small number of nuisance-driven false detections can distort vascular readouts, contaminate functional maps, and undermine confidence in interpretation. Tissue motion, phase instability, and aliased high-velocity flow therefore matter most when they survive ordinary clutter suppression and populate the extreme right tail of the detector output.

We address that problem with a localized matched-subspace detector-head framework. The historical borrowing from STAP is not a literal port of classical array-domain STAP. Instead, it is a shift toward local nuisance-aware detector design after ordinary residualization. The main practical result is that the simplest member of this framework, a band-limited matched-subspace score without local whitening, is the strongest fixed head on the prespecified held-out structural benchmark. A second, prespecified adaptive member keeps that score by default and turns on local whitening only when residual guard-band structure indicates localized clutter. A third, fully whitened member is retained for regimes where full covariance adaptation earns its extra complexity.

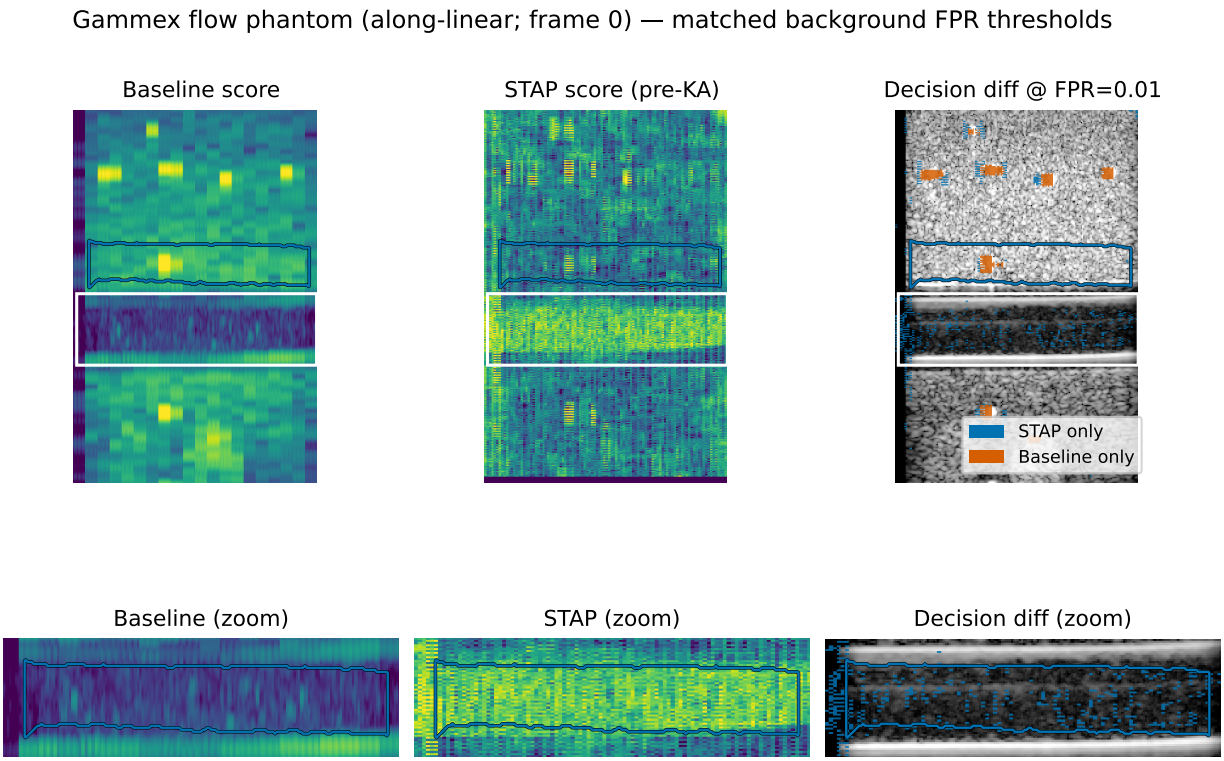

Figure 2: Leading structural-fidelity visual on a structurally labeled Gammex flow phantom acquisition from the open Doppler Twinkling Artifact database (labels derived from B-mode-only tube segmentation). Top row: baseline score map S_{base} (SVD band-pass clutter suppression followed by power Doppler), STAP pre-KA matched-subspace score $S_{\text{stap,pre}}$, and a matched-FPR decision-difference panel at a fixed background false-positive rate (10^{-2}; chosen here for visualization), showing pixels detected by STAP only versus baseline only. Bottom row: zoomed region of interest. The tube contour is overlaid for reference.

Instead of treating clutter suppression as the whole problem, we treat it as the first stage of a detection pipeline: a coarse residualizer removes the dominant clutter, and a second scoring stage asks whether the remaining signal is more consistent with parenchymal microvascular flow than with structured nuisance. This targets the spatially varying interference patterns that often remain after otherwise reasonable low-rank filtering and that disproportionately affect the low-FPR tail. Figure 1 previews the clearest real-data version of that story: on real in-vivo rat-brain IQ, the same-residual whitened matched-subspace specialist is more concentrated on the localization-derived vascular core and less contaminated by the immediately adjacent shell than conventional Doppler scoring. That real-IQ endpoint rewards concentration on the localization-derived vascular core under local whitening, whereas the SIMUS nuisance-control benchmark rewards fixed downstream robustness against broader structured nuisance; the winning head is therefore not the same in the two regimes.

This is not only an extreme-tail effect. On the representative Gammex frame in Figure 2, matching the background FPR at 10^{-2} yields zero lumen recovery for the baseline score, whereas the STAP score recovers about 10% of lumen pixels at the same matched background allowance. We therefore use the stricter 10^{-3}–10^{-4} operating points mainly as a stress test that exposes nuisance

sensitivity cleanly, not because the practical relevance begins only there.

We also study a conservative shrink-only regularization layer, but only as a bounded optional add-on: it can compress extreme-tail artifacts without score inflation, yet the main paper does not rely on it for its primary efficacy claims (details in the Supplementary Material).

What this paper contributes. The contribution of this paper is not a new clutter filter in isolation, but a localized matched-subspace detector-head framework for post-beamformed fUS: a nuisance-aware family centered on band-limited matched-subspace scoring, with a prespecified adaptive whitening switch and local covariance whitening where that extra complexity is justified.

Why this matters now. fUS is no longer only a preclinical proof-of-concept modality. It has already been used for intra-operative human functional mapping, human newborn functional imaging, bedside neonatal connectivity monitoring, adult cranial-window functional imaging, and emerging mobile human implementations through sonolucent implants [1, 8, 9, 20–22]. In that setting, the practical bottleneck is increasingly not whether some hemodynamic signal can be acquired at all, but whether the resulting maps remain trustworthy when rare nuisance structure or mild motion contaminates the extreme score tail. Our contribution is aimed at that narrower but clinically important problem: reducing nuisance leakage after ordinary clutter suppression so that downstream vascular and functional maps are easier to interpret at matched target sensitivity.

Contributions.

- A tile-local matched-subspace detector-head framework for fUS slow-time data, centered on a fixed band-limited matched-subspace head and extended by a prespecified adaptive whitening switch plus a bounded fully whitened specialist mode.
- Evidence that the fixed band-limited matched-subspace head is the strongest default on the prespecified held-out structural benchmark, together with clearer evidence about where the adaptive and fully whitened extensions do and do not add value.
- An evaluation framework that separates clutter suppression from the downstream scoring stage, includes strong low-rank and tensor baselines, and emphasizes strict low-FPR behavior rather than image appearance alone.
- Reproducible latency and deployment analyses that treat conditional execution explicitly as a changed method, not as an unconditional speedup, and that include fixed-calibration threshold-transfer checks on both the main SIMUS/PyMUST benchmark and the bounded PALA-backed ULM audit.
- A prespecified held-out SIMUS/PyMUST benchmark, including a held-out stress-frontier confirmation under cardiac-like pulsation and short ensembles, together with bounded audits on multiple open real-data resources, including a localization-derived structural-label check on real in-vivo rat-brain IQ, that support a concrete detector-family policy without over-claiming universal superiority.

Scope and claims. Claims are stratified by data regime and summarized in the evidence map of Table 2. Quantitative detection efficacy is limited to regimes with independent H_1/H_0 labels or a prespecified structural reference that is external to the short-window detector outputs being scored. Open in-vivo IQ and PD-only datasets are still used primarily for bounded robustness and translation audits, but the ULM section now also includes a localization-derived structural audit

reported explicitly as a bounded in-vivo vascular-fidelity endpoint rather than as a task-activation benchmark.

Organization. Section 2 defines the detection problem and score taxonomy. Section 3 presents the localized detector, with the fixed head first and the adaptive/fully whitened extensions second; the optional shrink-only regularizer is summarized briefly and left to the Supplementary Material for full details. Section 4 describes the prespecified operating settings and integrity controls. Section 5 reports results, followed by discussion and conclusion.

2 Background and Problem Formulation

Problem setup. For each spatial location (or tile-aggregated snapshot), let $x \in \mathbb{C}^{L_t}$ denote a slow-time vector. We consider

$$H_0 : \; x = c + n, \qquad H_1 : \; x = s + c + n,$$

where s is flow-consistent signal, c is structured clutter/motion, and n is residual noise. In practice, H_0 is spatially nonstationary and heavy-tailed, motivating local covariance adaptation and tail-focused evaluation.

Core notation and bands. We use band operators (P_o, P_f, P_g, P_a) for clutter/DC, flow, guard, and alias regions, respectively. For short ensembles, these are exact orthogonal DFT-bin projectors; for longer apertures they are leakage-controlled analysis operators and are treated as band-energy statistics with explicit sanity sentinels (Supplementary Material).

2.1 Related context

STAP and knowledge-aided STAP. Classical MVDR/LCMV/STAP and knowledge-aided variants shape covariances using external structure [48–51, 56, 57, 60]. In radar, KA is often expressed as covariance shaping/blending or knowledge-aided covariance estimation; here we adopt the KA spirit but implement it as a shrink-only, fail-safe score regularizer on top of a fixed detector statistic, with explicit conditions for inert behavior out of regime.

Ultrasound clutter suppression. Spatiotemporal SVD clutter filters are a standard tool in ultrafast Doppler and fUS [6, 7, 11, 12, 18]. Real-time feasibility has been studied via randomized/approximate decompositions and downsampling strategies [17], as well as GPU-parallel eigen-based clutter filtering frameworks [16]. Robust PCA, unfolded/learned RPCA, and tensor/HOSVD variants have also been studied for clutter suppression and motion robustness [26,27,40]. More generally, when raw channel/aperture data are available, aperture-domain multidimensional clutter filtering can provide additional degrees of freedom beyond beamformed slow-time processing [39]. We benchmark STAP against MC–SVD and established SVD-family clutter filters spanning adaptive-global SVD (spatial singular-vector similarity cutoffs), block-wise local SVD with fixed tilewise energy thresholds, and an adaptive-local hybrid that applies the same similarity-drop rule tilewise [12, 14], as well as RPCA and HOSVD baselines; conventional short-ensemble Doppler comparators are additionally reported on phantom data. These baselines are included to represent state-of-practice eigen/SVD clutter filtering beyond fixed-cutoff global Casorati SVD and to preempt the common critique that strict-tail gains are attributable only to localization or cutoff tuning rather

than to the detector formulation and tail operating discipline. We therefore benchmark against representative public ultrasound-native clutter-filter families, including adaptive-global SVD, local block-wise SVD, adaptive-local SVD, RPCA, HOSVD, and conventional Doppler score heads; our claim is superiority over those evaluated baseline families under the paper's low-FPR detector-head protocol, not blanket state-of-the-art superiority over every published pipeline.

What is new relative to Yu, Baranger, and Song. Eigen-based clutter filter design for ultrasound color flow imaging and block-wise/local adaptive clutter filters are closely related ultrasound-native approaches to covariance/eigenspace adaptation [13, 14, 16]. Song *et al.* demonstrate local clutter filtering in *in vivo* settings (e.g., kidney imaging), while Baranger *et al.* provide an adaptive singular-vector-similarity rank rule for ultrafast Doppler [12]. Our benchmark set includes both literature families directly and a tile-local hybrid that combines Song-style overlap-add localization with a Baranger-style adaptive rank rule. Many open fUS releases do not preserve raw high-PRF complex IQ suitable for labeled Doppler-side detector evaluation, so we use labeled k-Wave scenarios and structurally labeled phantoms for quantitative ROC claims and use *in vivo* data for robustness and translation audits. Our contribution differs in where the adaptation enters and how it is evaluated: Yu-, Baranger-, and Song-style methods adapt the *clutter filter* itself and then typically read out power Doppler or a closely related conventional score, whereas we keep the residual cube fixed and change the *detector head*. Specifically, we compute a GLRT-motivated matched-subspace score on tile-local slow-time snapshots with explicit Doppler band geometry (flow/guard/alias), add a prespecified adaptive whitening switch only when guard-band evidence indicates localized clutter, evaluate same-residual score changes separately from upstream filtering, and report the resulting behavior in the strict low-FPR regime where background-tail failures dominate interpretation. This is therefore not another local SVD clutter filter with radar terminology on top; it is a downstream detector-head framework for post-residual fUS.

ULM pipelines and motion correction. Ultrafast ultrasound localization microscopy (ULM) achieves deep super-resolution vascular imaging by localizing sparse microbubble echoes over high-PRF ensembles [35]. Robust clutter suppression and motion correction are central to ULM pipelines, including subwavelength motion correction tailored to ultrafast acquisitions [36]. We do not claim improved microbubble localization performance here; rather, we use a public kHz IQ ULM benchmark dataset to report label-free stability and motion robustness of Doppler-side score maps under controlled motion corruption, illustrating how a tail-stable detector can complement upstream motion correction and downstream localization stages.

Twinkling artifact and Doppler phantoms. The Doppler "twinkling artifact" has been documented as a persistent color-flow phenomenon on calcifications and other strong reflectors, and is widely used as a diagnostic benchmark for Doppler processing chains. Leonov *et al.* provide an open-access database of raw beamformer signals with phantoms including a flow-through Gammex phantom and calculi/metal/plastic objects [38]. We use these data to (i) obtain a structurally labeled phantom ROC via B-mode-only tube segmentation and (ii) construct an artifact-heavy KA hygiene regime where the prior penalty is expected to be conservative and auditable.

fUS detection and hemodynamic analysis. GLM-based detection and atlas-driven ROI analysis are common in hemodynamic fUS [43, 44]. In this paper we use those analyses as downstream readouts and translation audits; extending the detector to a fully dedicated hemodynamic scoring variant is future work rather than a claimed contribution here.

Symbol / name	What it measures	Used for	Data requirement
S_{base}	baseline ROC score map on X_{base} (PD power; higher = more flow-like)	baseline ROC / low-FPR tables	IQ
$S_{\text{stap,pre}}$	pre-KA matched-subspace ratio STAP score map	STAP ROC / low-FPR tables	IQ + local covariance
$S_{\text{stap,post}}$	post-KA shrink-only score map $wS_{\text{stap,pre}}$	KA behavior audits	IQ + feature map
$\text{PD}^{\uparrow}$ (display PD), masks	PD display/proxy map and derived masks	visualization and deterministic preprocessing (gating/protected sets)	IQ (or PD-only proxy mode)

Table 1: Score/map taxonomy used throughout. ROC score maps are kept separate from display/proxy maps used only for visualization and deterministic preprocessing. Display/proxy maps are never used to compute ROC/TPR/FPR (only to define gating/protected-set masks, which are audited separately).

2.2 Baseline clutter suppression and score conventions

Beamformed slow-time data. Let $X[t, y, x] \in \mathbb{C}$ denote a beamformed slow-time IQ movie over $t = 0, \ldots, T-1$ and spatial indices (y, x). We first apply a baseline clutter-suppression operator (e.g. motion-compensated SVD), producing a residual cube $X_{\text{base}}[t, y, x]$ used for both baseline scoring and as the input to STAP.

Power Doppler proxy and baseline scores. From the baseline-filtered cube we define a conventional power Doppler (PD) brightness map used only as a display/proxy map for deterministic preprocessing (e.g. proxy masks) and visualization,

$$\text{PD}^{\uparrow}(y, x) = \frac{1}{T} \sum_{t=0}^{T-1} |X_{\text{base}}(t, y, x)|^2 .$$

For quantitative evaluation and thresholding we use *flow-evidence* score maps where higher values indicate stronger evidence of flow, and we calibrate thresholds on the right tail of these score distributions. We denote the baseline ROC score map as $S_{\text{base}}(y, x)$ (power Doppler on X_{base}). For short-ensemble regimes we additionally report auxiliary, conventional baseline comparators (e.g. log-power Doppler and Kasai lag-1 magnitude [15]) to avoid baseline-score idiosyncrasies dominating conclusions (Notation). Note: $\text{PD}^{\uparrow}$ and S_{base} use the same underlying power statistic (up to fixed scaling); we keep separate names to make it explicit when PD is used only as a display/proxy map (masks/visuals) versus as a ROC score map.

2.3 Score taxonomy

Unless explicitly stated, the paper discusses three pre-KA detector variants built from the same band-limited score family: a fixed matched-subspace head without whitening, a prespecified adaptive variant that promotes selected pixels onto a whitened branch when guard-band evidence indicates clutter, and a fully whitened covariance-adaptive variant. To keep the notation manageable,

the remainder of the paper refers to these simply as the fixed, adaptive, and fully whitened detector modes. KA is analyzed separately as an optional, shrink-only post-processing regularizer.

3 Method

What changes relative to SVD + PD. Relative to a conventional ultrafast Doppler pipeline, the proposed detector family changes the final scoring stage rather than replacing upstream clutter suppression. After forming a residual slow-time signal, we score its energy inside a prespecified flow subspace rather than using only aggregate Doppler power. The simplest fixed head uses that score without local whitening, a prespecified adaptive variant promotes clutter-heavy pixels onto a whitened branch only when guard-band evidence warrants it, and a fully whitened variant is retained for regimes where stronger local adaptation is beneficial. The goal is not simply to increase vessel conspicuity, but to reduce nuisance-driven high scores in operating regions where false positives matter.

Radar terms translated into ultrasound terms. We retain a few radar-inspired abbreviations because they compactly describe the design logic, but in this paper they refer to specific post-beamformed ultrasound operations. For readers coming from ultrasound rather than radar, the fixed head can be read simply as a local band-limited flow detector, the adaptive branch as tile-local covariance whitening that turns on only in clutter-heavy regions, and the optional penalty layer as a conservative vessel-proxy masking rule that can only shrink scores. In radar terms, the fully whitened branch with the optional penalty layer is closest to knowledge-aided STAP; in ultrasound terms, it is a tile-local covariance-adaptive matched filter with an optional heuristic penalty map. "STAP" refers to tile-local covariance adaptation in slow time, i.e. optional local whitening of beamformed residual snapshots before matched-subspace scoring. "CFAR" refers to how operating points are set, namely empirical threshold calibration on labeled background or nuisance tails rather than an analytic radar-style false-alarm law. "KA" refers to a heuristic score penalty based on auxiliary band-energy features and vessel-proxy masks, implemented here only as a shrink-only regularizer that can suppress high-risk tiles but never amplify them. These are therefore ultrasound-specific detector components, not a literal port of classical radar processing blocks.

Pipeline overview.

1. Baseline clutter suppression (e.g. MC–SVD) produces a residual cube X_{base} and baseline score $S_{\text{base}}(y, x)$.
2. For each tile Ω_i: form slow-time snapshots $\{x_{i,k}\}_{k=1}^{K}$ via spatial sampling and, optionally, a sliding-window temporal embedding (Hankel matrix construction), estimate conditioned covariance R_i, and either keep the raw snapshots or whiten them as $y_{i,k} = R_i^{-1/2} x_{i,k}$.
3. Compute a matched-subspace ratio score (ACE/MSD-style) on the chosen tile representation, e.g. $S_{\text{stap,pre}}(i) = \frac{1}{K}\sum_k \frac{y_{i,k}^{\text{H}} P_f\, y_{i,k}}{y_{i,k}^{\text{H}}(I-P_f)\, y_{i,k} + \rho\, y_{i,k}^{\text{H}} y_{i,k} + \epsilon}$, and aggregate over overlapping tiles to obtain $S_{\text{stap,pre}}(y, x)$. Setting $R_i = I$ recovers the simpler band-limited matched-subspace fixed head.
4. If the adaptive variant is used, retain the fixed head by default and promote pixels onto the whitened branch only where a prespecified guard-band clutter feature exceeds a fixed threshold.

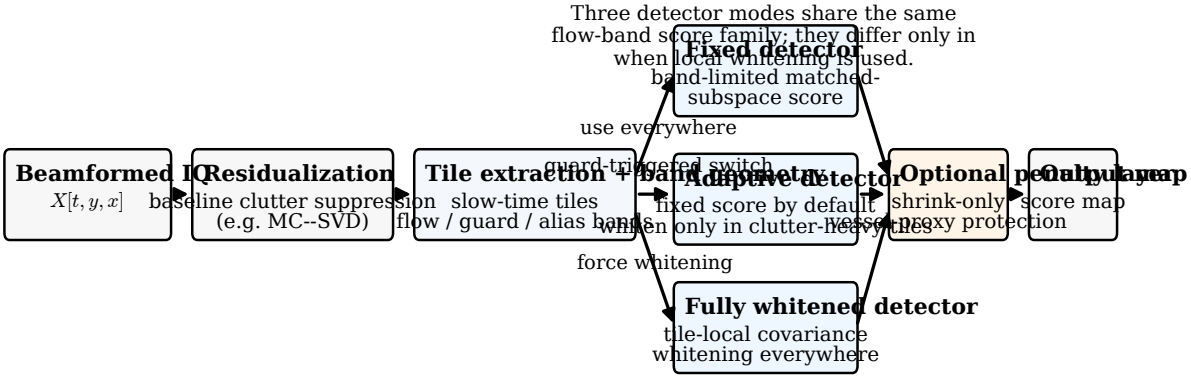

Figure 3: Method overview. Beamformed IQ first undergoes ordinary residualization, after which the detector operates tilewise in slow time. The fixed detector uses the band-limited matched-subspace score everywhere. The adaptive detector uses that same score by default and switches to a whitened branch only in tiles with strong guard-band evidence of localized clutter. The fully whitened detector applies whitening everywhere. An optional shrink-only penalty layer can then downweight high-risk regions while leaving protected vessel-proxy regions unchanged.

5. If the optional penalty layer is enabled: compute a deterministic, shrink-only weight $w(i) \in (0, 1]$ from auxiliary band-energy statistics and vessel-proxy support features (details in the Supplementary Material), preserve a protected do-not-penalize set (scores unchanged), and output $S_{\text{stap,post}}(y, x) = w\, S_{\text{stap,pre}}(y, x)$. Else, $S_{\text{stap,post}}(y, x) = S_{\text{stap,pre}}(y, x)$.

In practical terms, the method asks a local question after ordinary clutter suppression: within a small spatial neighborhood, does the remaining slow-time activity look more like microvascular flow or more like locally structured nuisance? A STAP-only ablation on phantom data is provided in Supplementary Material.

Three detector variants. Let $S_0(i)$ denote the non-whitened band-limited matched-subspace score on tile i and let $S_w(i)$ denote the fully whitened Huber-STAP score computed from the same band geometry. The paper studies three concrete detector variants. The fixed detector uses $S_0(i)$ everywhere. The prespecified adaptive detector uses

$$S(i) = \begin{cases} S_w(i), & r_g^{\text{base}}(i) \geq \tau_g, \\ S_0(i), & r_g^{\text{base}}(i) < \tau_g, \end{cases}$$

where $r_g^{\text{base}}(i)$ is a baseline guard-band energy fraction and τ_g is fixed once and then held fixed across all reported experiments. The fully whitened detector uses $S_w(i)$ everywhere. In words, the adaptive detector keeps the simpler score in clean tiles and only turns on full whitening where the residual itself shows guard-band evidence of localized clutter.

3.1 Localized detector core and Doppler band geometry

Tile-based detector family. To mitigate the degraded tail sensitivity of global low-rank filters in high-clutter regimes, we study a localized matched-subspace detector family on overlapping spatial tiles. Its primary member keeps the same flow-band score without local whitening ($R_i = I$); its adaptive member uses that same score by default and promotes selected tiles onto a whitened branch; its fully whitened member adds a STAP-inspired temporal core in which adaptation is in slow time and spatial localization enters through tile-local covariance adaptation. Let Ω_i denote tile i and L_t the slow-time aperture. From X_{base} we construct a set of slow-time training snapshots $\{x_{i,k} \in \mathbb{C}^{L_t}\}_{k=1}^{K}$ for each tile via spatial sampling and, when applicable, a sliding-window temporal embedding (Hankel construction) in slow time. Here K denotes the number of training snapshots used to estimate the tile covariance; in the fully Hankelized case, each pixel contributes $N = T - L_t + 1$ snapshots and pooling across Ω_i yields $K = N|\Omega_i|$ (or a subsampled/capped variant). We estimate a local covariance

$$\widehat{R}_i := \frac{1}{K} \sum_{k=1}^{K} x_{i,k} x_{i,k}^{\mathrm{H}},$$

apply conditioning/diagonal loading to obtain R_i, and whiten snapshots as $y_{i,k} := R_i^{-1/2} x_{i,k}$.

Training support and Hankel dependence. In the fully Hankelized case, each pixel in Ω_i contributes $N = T - L_t + 1$ overlapping slow-time snapshots; pooling across the $|\Omega_i|$ pixels yields $K = N|\Omega_i|$ tile-local training vectors. These Hankelized snapshots are temporally correlated due to overlap, so K should not be interpreted as an independent-sample count. Our robust covariance estimators do not require independence, but the effective support can be significantly smaller than K; we therefore optionally subsample Hankel columns (stride) and cap the number of columns used for training to bound computation and avoid overweighting highly correlated samples (fixed per regime; Supplementary Material).

Robust covariance estimation and conditioning for the fully whitened mode. Whenever the fully whitened member of the detector family is used, we whiten with a regularized covariance R_i rather than the raw SCM $\widehat{R}_i$. The prespecified stress-test tables in this paper use a Tyler-type M-estimator on the pooled tile snapshots, iterating

$$R_i^{(n+1)} \leftarrow \frac{L_t}{K} \sum_{k=1}^{K} \frac{x_{i,k} x_{i,k}^{\mathrm{H}}}{x_{i,k}^{\mathrm{H}} (R_i^{(n)})^{-1} x_{i,k}}, \qquad R_i^{(n+1)} \leftarrow \frac{L_t}{\mathrm{tr}(R_i^{(n+1)})} R_i^{(n+1)},$$

with at most 25 iterations and early stopping when $\|R^{(n+1)} - R^{(n)}\|_F / \|R^{(n)}\|_F < 10^{-4}$ (latency-only experiments may cap iterations; Supplementary Material). We then apply deterministic conditioning (shrinkage toward μI and diagonal loading $\lambda \mu I$, where $\mu = \mathrm{tr}(R)/L_t$ and λ is fixed per regime) to enforce target condition numbers. In very short-ensemble regimes we may fall back to a Huber IRLS covariance estimate for numerical stability (Supplementary Material). When the whitened member of the detector family is used, we use a single robust recipe based on Huber covariance estimation with trimmed high-energy training support and stronger loading. We refer to that prespecified recipe as the *Huber-whitened STAP variant*; exact numerical settings remain in the reproducibility appendix so the main text can focus on which member of the family is preferred in which regime.

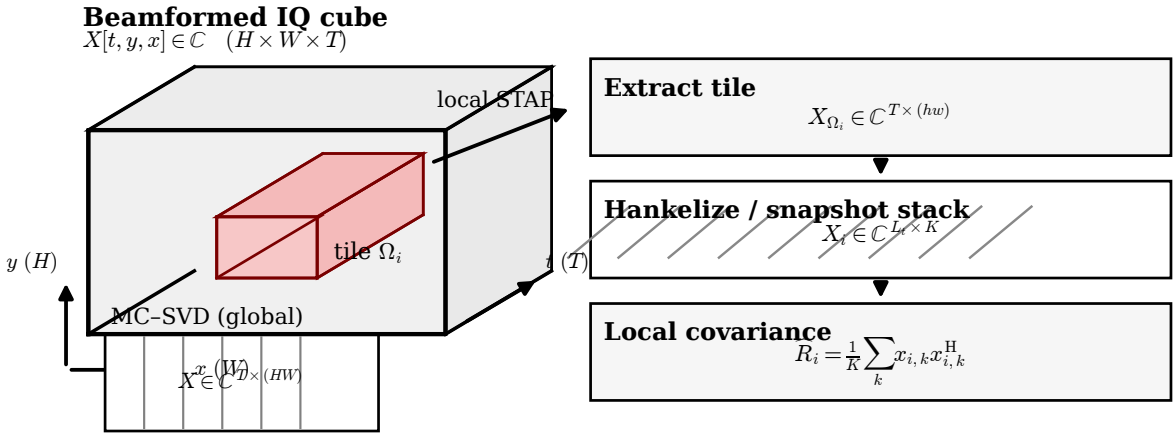

Figure 4: Schematic of sliding-window temporal embedding (Hankel matrix construction) and local covariance estimation. A localized spatial tile Ω_i is extracted from the beamformed slow-time data cube $X[t, y, x] \in \mathbb{C}$ $(H \times W \times T)$. Overlapping slow-time snapshots of length L_t are stacked (optionally Hankelized) to estimate the localized covariance matrix $\widehat{R}_i$ used only by the advanced whitened mode. Unlike global low-rank filters (e.g. MC–SVD) that operate on a global Casorati matrix, this localized formulation preserves spatial-spectral heterogeneity of microvascular flow and highly structured tissue clutter.

Terminology. We use "STAP-inspired" to denote the broader detector framework introduced here: localized matched-subspace scoring after ordinary residualization, together with a prespecified adaptive rule and an optional fully whitened covariance-adaptive variant in slow time. The primary fixed detector in this paper is the non-whitened band-limited matched-subspace score. The adaptive variant uses that score by default and only promotes tiles onto the whitened branch when guard-band clutter evidence is high. The fully whitened member adds local covariance adaptation, whitening, and the same subspace-based scoring when that extra complexity is justified. Unlike classical radar STAP, the method does not exploit array-channel degrees of freedom or an explicit space–time steering manifold. The analogy is therefore architectural rather than literal.

Figure 4 illustrates the local tile extraction and optional sliding-window temporal embedding (Hankel construction) used to form $\widehat{R}_i$ and contrasts it with the global Casorati stacking used by MC–SVD.

Band operators and matched-subspace STAP score. We partition the Doppler spectrum using band projectors (P_o, P_f, P_g, P_a) (Notation). For short ensembles, these are exact, orthogonal DFT-bin projectors with $P_o + P_f + P_g + P_a = I$; for longer apertures they are leakage-controlled operators that are approximately orthogonal and are treated as band-energy statistics. We define a matched-subspace STAP test statistic by flow-subspace energy on whitened snapshots, e.g.,

$$S_{\mathrm{stap,pre}}(i) := \frac{1}{K} \sum_{k=1}^{K} \frac{y_{i,k}^{\mathrm{H}} P_f \, y_{i,k}}{y_{i,k}^{\mathrm{H}} (I - P_f) \, y_{i,k} + \rho \, y_{i,k}^{\mathrm{H}} y_{i,k} + \epsilon},$$

and form a pixel-level score map $S_{\text{stap,pre}}(y, x)$ by overlap-add aggregation of the contributing tile statistics. This pre-KA score is the primary STAP detector output used for quantitative reporting unless explicitly stated otherwise.

Regularization and self-normalization avoidance. If we whitened using the exact sample covariance formed from the same snapshots being scored, the statistic collapses: the whitened sample covariance becomes (approximately) identity and averaged subspace energies concentrate near constants (e.g. $\mathrm{tr}(P_f)$), destroying discrimination. We avoid this by whitening with a *regularized* covariance R_i (robust estimation plus shrinkage/diagonal loading), so that typically $R_i \neq \widehat{R}_i$ and the matched-subspace energy remains informative. We do not rely on an analytic null distribution for CFAR operation; operating points are empirically calibrated on labeled H_0 sets under the prespecified protocol of Section 4.1.

Matched-subspace statistic as a whitened subspace detector. Fix tile i and consider a single slow-time snapshot $x \in \mathbb{C}^{L_t}$. Under a Gaussian clutter-plus-noise model,

$$H_0 : \; x \sim \mathcal{CN}(0, R_i), \qquad H_1 : \; x = A_f s + n, \quad s \sim \mathcal{CN}(0, \sigma_s^2 I), \quad n \sim \mathcal{CN}(0, R_i),$$

where the columns of A_f span the flow subspace (equivalently the range of P_f). After whitening $y = R_i^{-1/2} x$, a matched-subspace detector based on whitened flow-subspace energy arises naturally under this colored-Gaussian model:

$$T(x) = x^{\mathrm{H}} R_i^{-1/2} P_f R_i^{-1/2} x = y^{\mathrm{H}} P_f \, y,$$

which is the standard whitened subspace-energy statistic for this setting. [58,59] In implementation we report a self-normalized ACE-like ratio score that normalizes flow-subspace energy by the orthogonal complement (and a small ridge). It is motivated by the same whitened subspace geometry as $T(x)$, but it is not identical to the basic GLRT energy statistic; we use it because it behaves better numerically and yields more stable thresholding in practice. This self-normalized statistic is the pre-KA detector score $S_{\text{stap,pre}}(i)$.

3.2 Optional shrink-only regularization (brief main-text summary)

The main detector results in this paper are all carried by the pre-regularization detector heads above. We nevertheless study a bounded optional shrink-only regularizer because structured artifacts can still dominate extreme-tail operating points in some regimes. The mapping is deterministic, conservative, and uses auxiliary band-energy and proxy-support statistics only as *side information*; proxies are never used to compute ROC/TPR/FPR (only to define protected/candidate sets and compute masks, which are audited separately). Full construction details and safety contracts are left to the Supplementary Material.

Regularizer at a glance.

- **When it activates:** Only when deterministic sanity/support checks pass (valid band discretization; non-flat risk features; sufficient proxy support). Otherwise it is inert ($w \equiv 1$).
- **What it changes:** Applies a bounded, shrink-only weight $w(i) \in (0, 1]$ on an explicitly defined candidate set while preserving a protected do-not-penalize set (scores unchanged); the post-KA score is $S_{\text{stap,post}} = w\, S_{\text{stap,pre}}$.
- **What it cannot do:** It cannot increase any pixel score (no amplification) and proxies are not used for ROC/TPR/FPR reporting; any apparent operating-point uplift reflects background-tail compression under shrink-only mapping.
- **How it fails safely:** If out of regime, it defaults to "off" ($w \equiv 1$) and logs an explicit reason label, reducing to the pre-regularization detector.

Core KA feature definitions. KA is driven by a small set of deterministic side-information features computed from the same tile-local slow-time vector z_i and Doppler-band projectors (P_o, P_f, P_g, P_a) used by the detector. We define

$$E_f(i) = z_i^{\mathrm{H}} P_f z_i, \qquad E_g(i) = z_i^{\mathrm{H}} P_g z_i, \qquad E_a(i) = z_i^{\mathrm{H}} P_a z_i,$$

with total in-band energy $E_{\text{tot}}(i) = z_i^{\mathrm{H}} z_i + \epsilon$. The main KA risk statistic is the alias-to-flow ratio

$$m_{\text{alias}}(i) = \log \frac{E_a(i) + \epsilon}{E_f(i) + \epsilon},$$

which is large when a tile is dominated by aliased or high-velocity nuisance rather than the prescribed flow band. We also monitor the guard-band fraction

$$r_g(i) = \frac{E_g(i)}{E_{\text{tot}}(i)},$$

and a deterministic flow-support proxy $c_f(i) \in [0, 1]$ to define conservative background-like and flow-like proxy sets

$$\mathcal{T}_{\text{bg}} := \{i : c_f(i) \le c_{\text{bg}}\}, \qquad \mathcal{T}_{\text{flow}} := \{i : c_f(i) \ge c_{\text{flow}}\}.$$

These quantities are used only to decide whether shrinkage is warranted and where it may be applied; they are not used as class labels for ROC reporting.

Shrink-only mapping (guarantee). We implement the optional penalty layer (knowledge aid in radar terminology) via a bounded weight $w(i) \in (0, 1]$ applied in score space:

$$S_{\text{stap,post}}(i) = w(i)\, S_{\text{stap,pre}}(i), \qquad w(i) \in (0, 1].$$

The Supplementary Material gives the full construction of $w(i)$, including protected sets, regime logic, deterministic sentinels, and the reciprocal scale-map convention used for some exported audit files.

Lemma 1 (Shrink-only monotonicity)**.** *Let $w(i) \in (0, 1]$ and define $S_{\text{stap,post}}(i) = w(i)\, S_{\text{stap,pre}}(i)$. For any threshold τ,*

$$\{S_{\text{stap,post}} \ge \tau\} \subseteq \{S_{\text{stap,pre}} \ge \tau\}.$$

Consequently, on any subset A (e.g. H_0), the tail probability cannot increase at fixed τ:

$$\mathbb{P}\{S_{\text{stap,post}} \ge \tau \mid A\} \le \mathbb{P}\{S_{\text{stap,pre}} \ge \tau \mid A\}.$$

Proof. If $S_{\text{stap,post}}(i) \geq \tau$, then $S_{\text{stap,pre}}(i) = S_{\text{stap,post}}(i)/w(i) \geq \tau$ since $w(i) \leq 1$. □

When reporting strict low-FPR operating points where τ is reselected post-KA to satisfy $\mathbb{P}\{S_{\text{stap,post}} \geq \tau \mid H_0\} = \alpha$, changes in TPR reflect background-tail compression despite scores never increasing.

3.3 Algorithmic integrity and deterministic safeguards

To make the architecture auditable and to avoid leakage or discretization artifacts masquerading as performance gains, we enforce deterministic, label-free safeguards:

- **Conditional execution as a decision-rule variant.** To reduce latency, STAP can be skipped outside a flow-proxy region and replaced by the baseline score there. Because this changes the overall decision rule, any conditional-execution reporting is treated as a distinct algorithmic variant and is accompanied by leakage controls (same-window vs. disjoint-window vs. randomized proxy masks). Unless otherwise stated, conditional execution uses *disjoint-window* proxy gating for detection reporting.
- **Short-ensemble band discretization sanity.** For extremely short ensembles (e.g. $N \in \{5, 9, 17\}$), the coarse frequency resolution $\Delta f = \text{PRF}/N$ can cause requested bands to collapse or overlap after discretization. If the discretized bands are empty, DC-only, or overlapping, KA is disabled ($w \equiv 1$) with an explicit reason label.
- **Proxy-mask circularity control.** Proxy maps are never used to compute ROC/TPR/FPR; they are used only to define compute masks (conditional execution) or protected/candidate sets (KA), which are audited separately. Quantitative ROC claims are restricted to regimes with independent H_1/H_0 labels (simulator truth or B-mode structural geometry).

For auditability, we log explicit reason labels so that "KA off" ($w \equiv 1$) is an expected success mode under flat or non-actionable evidence. Supplementary Material provides modality-agnostic sufficient conditions and additional sentinels used to interpret when the KA prior should be active or inert.

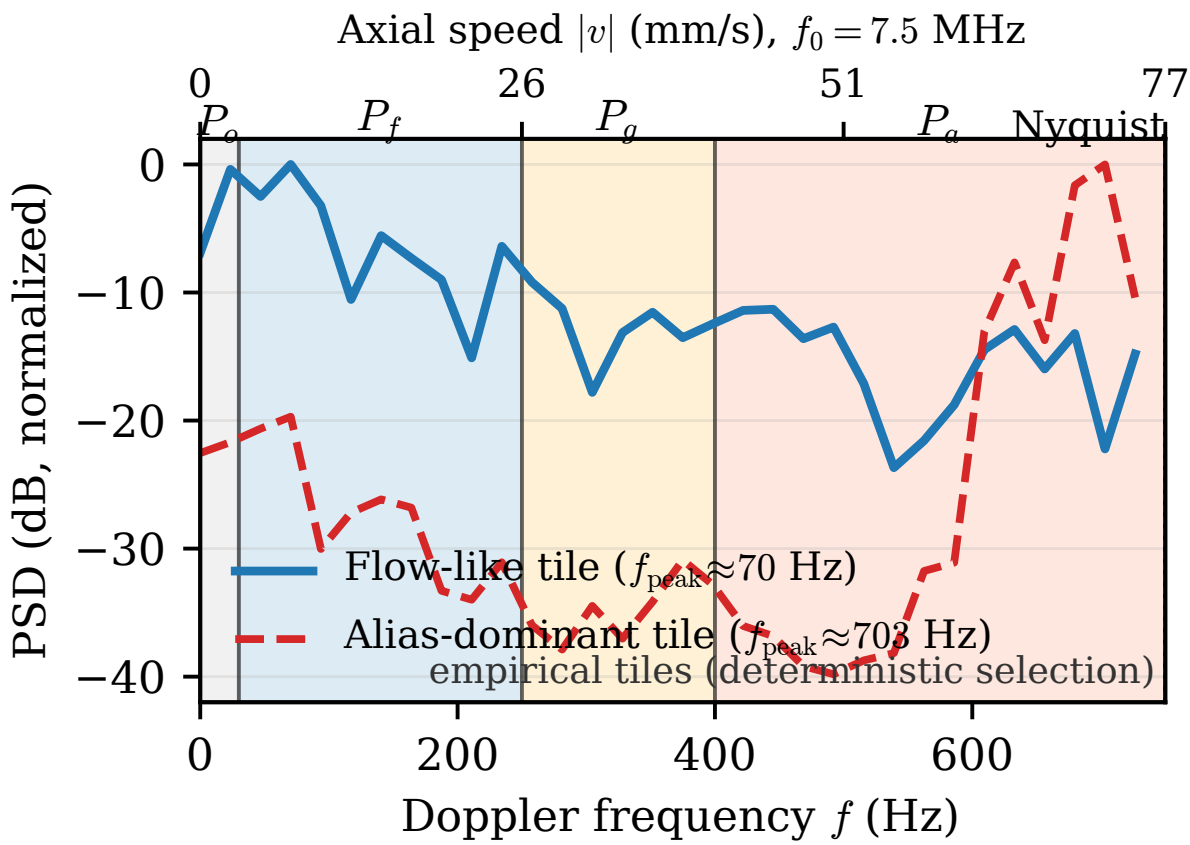

Figure 5: Doppler power spectral density (PSD) partitioning and kinematic band geometry (empirical example). Curves show one-sided PSDs of the tile-mean slow-time signal on the alias-dominant labeled brain simulation (PRF = 1500 Hz): a flow-like tile (peak in P_f) and an alias-dominant tile (peak in P_a). The Nyquist interval is deterministically partitioned into DC/high-amplitude clutter (P_o), microvascular flow (P_f), a transition guard band (P_g), and aliased high-velocity artifacts or vibration (P_a). Auxiliary band-energy descriptors (e.g. alias-to-flow ratios) are used for interpretation and, optionally, for shrink-only tail regularization (Supplementary Material).

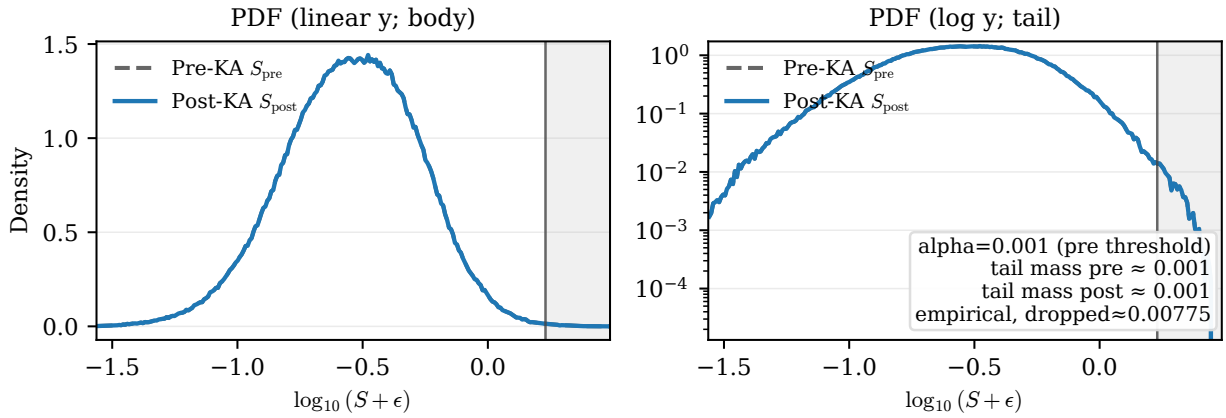

Figure 6: Empirical statistical effect of the strictly bounded, shrink-only penalty on a heavy-tailed background-proxy score distribution from an artificial-calculi acquisition in the open Doppler Twinkling Artifact database. The pre-KA matched-subspace score $S_{\text{stap,pre}}$ exhibits a heavy right tail characteristic of structured artifacts. Applying the non-increasing weight $w(i) \in (0, 1]$ yields the post-regularization score $S_{\text{stap,post}} = w(i)S_{\text{stap,pre}}$, compressing the background tail while preserving an explicit do-not-penalize protected set (scores unchanged on high-confidence flow locations). The vertical line indicates the pre-KA threshold τ_α chosen so that $\mathbb{P}\{S_{\text{stap,pre}} \geq \tau_\alpha \mid \text{bg proxy}\} = \alpha$ (here $\alpha = 10^{-3}$); the shaded region highlights the corresponding right tail, and the post-KA tail mass is evaluated at the same τ_α.

4 Experimental Design

4.1 Prespecified operating protocol and operating-point reporting

All processing settings (tile geometry, aperture, band definitions, conditioning targets, and prior thresholds) are fixed for each regime before held-out results are reported. Detection is evaluated by thresholding the high-score tail (right tail) at fixed FPR targets:

$$\mathbb{P}\{S \geq \tau_\alpha \mid H_0\} = \alpha, \qquad \mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}.$$

For each experiment, we report the supported floor $fpr_{\min} = 1/n_{\mathrm{neg}}$ and bootstrap uncertainty at the natural unit of variation (windows or frames, depending on regime). These choices enforce a strict low-FPR operating discipline and avoid over-interpreting unsupported tail points. We emphasize $\alpha = 10^{-3}$–10^{-4} because those operating points expose nuisance-tail failure modes cleanly, not because routine fUS use must literally operate at those exact values. The same map-trustworthiness problem is already visible at a conventional matched background FPR of 10^{-2} in the worked Gammex example of Figure 2, where the baseline thresholded map is dominated by off-lumen detections while the matched-subspace score recovers lumen-consistent structure.

Threshold calibration and deployment. Unless otherwise stated, τ_α is calibrated per window (or per cine sequence) on that unit's negative set to report $\mathrm{TPR}(\alpha)$ at the intended operating point; in labeled regimes, the negative set is available for *exact* calibration, while in deployment it would be estimated via a CFAR-style training procedure or from a fixed calibration set. This per-window protocol should therefore not be read as requiring oracle per-window tuning in deployment. To keep that distinction explicit, the Results section now includes two non-retrospective transfer checks in the main reader path: a SIMUS/PyMUST fixed-calibration audit in which thresholds are learned on separate calibration seeds and then applied unchanged to held-out evaluation seeds, and a complementary block-split fixed-calibration audit on the PALA-backed ULM structural endpoint.

How the prespecified defaults were chosen. The numerical settings in this pipeline do not all play the same role, and they were not selected in the same way. Band edges are tied to acquisition physics: PRF fixes the discrete Doppler grid and Nyquist limit, while the flow/guard/alias partitions are chosen to separate slow parenchymal microvascular content from transition-band and aliased high-velocity nuisance on that grid. Tile geometry and L_t are locality/support choices: they are chosen once per regime to preserve spatial adaptation while retaining enough slow-time support for stable covariance estimation, then held fixed. Conditioning parameters (for the fully whitened mode) are chosen to keep the robust covariance well behaved on development audits rather than to maximize held-out task scores. The adaptive threshold τ_g is a single frozen guard-energy cutoff chosen to preserve the fixed head in cleaner regimes while turning on whitening only when guard-band clutter evidence is clearly non-negligible. The KA proxy thresholds are deliberately coarse rather than finely optimized: $c_{\mathrm{bg}} = 0.05$ means "essentially background-like tile coverage," $c_{\mathrm{flow}} = 0.20$ means "substantial flow-support coverage," $q_{\mathrm{risk}} = 0.90$ defines a conservative upper-tail risk cutoff, and the p_{shrink} bounds are safety sentinels on how much of the candidate set may be penalized. A supplementary stability audit perturbs the finalized τ_g and c_{flow} thresholds by $\pm 10\%$ and shows that the detector family does not qualitatively collapse under those changes (Supplementary Material).

4.2 Integrity checks and deterministic sentinels

This work is a reproducible processing pipeline (decode → baseline → detector score → optional shrink-only penalty layer). Several practical pathologies can invalidate tail-rate reporting or spuriously activate the penalty layer (e.g. missing acquisition metadata, coarse short-ensemble band discretization, proxy-mask circularity, or compute-gating leakage). We therefore enforce deterministic, label-free integrity checks that either (i) record provenance and proceed conservatively or (ii) disable the penalty layer ($w \equiv 1$) with an explicit reason label. A concise checklist (and where each safeguard is logged) is provided in Supplementary Material.

4.3 Regimes and claim scope

Claims are stratified by regime so that quantitative detection results remain clearly separated from label-free robustness or translation audits. Table 2 also makes the role of open datasets explicit, because the real-data story in this paper is intentionally bounded but not singular.

- **Labeled brain simulation scenarios (k-Wave).** These are controlled stress tests for strict-tail behavior. Across the open-skull, alias-dominant, and structured-clutter scenarios, the prespecified STAP score retains measurable sensitivity at ultra-low FPR, with the strongest gains in the alias-dominant and structured-clutter settings.
- **Phantom acquisitions (structurally labeled Gammex flow phantom).** These provide independent structure-based labels on real scanner data. We derive B-mode-only tube masks, compare conventional short-ensemble Doppler scores with STAP-family scores, and report structural low-FPR operating points and within-ensemble motion ladders.
- **Prespecified held-out SIMUS/PyMUST benchmark.** This is the main translational benchmark. It uses the SIMUS/PyMUST forward model, but the evaluated settings are narrowed using real-data fUS use cases and observed operating characteristics rather than generic simulator coverage alone [8, 9, 30, 41–43]. We report held-out structural and functional performance for the prespecified mobile setting and intra-operative parenchymal setting, together with a held-out stress-frontier confirmation on the mobile setting under cardiac-like pulsation and short ensembles.
- **ULM Zenodo 7883227 (real kHz IQ).** This dataset is used for motion-robustness analyses on real IQ, with controlled rigid and nonrigid motion injection and explicit fairness checks. We also report a bounded structural-label audit based on the published PALA super-resolution vascular rendering registered back onto the IQ grid, using a vessel-core versus perivascular-shell endpoint rather than a functional-task label.
- **Shin rat-brain real-IQ sequence.** This open real-IQ dataset provides a second label-free motion-stress evaluation under a prespecified low-frequency operating setting. It is used as corroborating real-IQ evidence rather than as a task ROC benchmark.
- **Supplementary analyses.** Additional label-free audits (e.g. PD-only whole-brain fUS) and diagnostic panels are provided in Supplementary Material.

5 Results

Results roadmap. We begin with the prespecified held-out SIMUS/PyMUST moving-scatterer benchmark, because it establishes the main translational result of the paper: the band-limited matched-subspace detector is the strongest default downstream head on the held-out structural

Regime	Open-data status / data type	Independent labels?	What it supports
Labeled brain simulation scenarios (k-Wave)	project-generated simulated IQ	yes (simulator truth)	controlled strict-tail mechanism and latency stress tests
Twinkling database: Gammex + calculi	open real scanner phantom Doppler IQ	Gammex: yes (B-mode structure); calculi: no	structural ROC on real data, phantom motion ladders, and KA hygiene
Prespecified held-out SIMUS/PyMUST benchmark	open SIMUS/PyMUST forward model with real-data anchoring	yes (simulated label masks)	main held-out structural and functional translational benchmark, held-out stress-frontier confirmation under cardiac-like pulsation and short ensembles, and separate-bank fixed-calibration transfer on held-out seeds
ULM Zenodo 7883227	open real kHz IQ	bounded structural labels from a published localization-derived vascular reference	real-IQ motion robustness, same-residual score audit, a bounded structural-label vascular-fidelity audit on real in-vivo brain IQ, and complementary block-split fixed-calibration transfer
Shin rat-brain IQ sequence	open real kHz IQ	no independent task labels	second real-IQ corroboration under controlled residual-motion stress

Table 2: Evidence ladder and claims map. Quantitative detection performance is reported across three labeled tiers: controlled brain simulation, structurally labeled phantom data from the open Twinkling database, and one bounded in-vivo structural-label audit on ULM 7883227 using a published localization-derived vascular reference. The prespecified held-out SIMUS/PyMUST moving-scatterer benchmark provides the main translational stress-test tier. The other real-IQ sections remain explicitly bounded as robustness, same-residual, or corroborative audits rather than as broad task ROC benchmarks.

task. We then report the supporting labeled brain simulation stress tests at ultra-low FPR, the structurally labeled Gammex phantom results, and finally the open real-IQ evaluations (ULM 7883227 and the Shin rat-brain sequence), including a bounded localization-derived structural-label audit on real in-vivo brain IQ. Additional dataset evaluations and diagnostic panels are provided in Supplementary Material.

5.1 Prespecified held-out SIMUS/PyMUST benchmark: what is being measured

This moving-scatterer benchmark is intentionally narrower than a generic SIMUS parameter sweep. It retains the SIMUS/PyMUST moving-scatterer forward model [41, 42], but the evaluated settings are chosen to reflect whole-brain, cranial-window, mobile, and intra-operative fUS use cases rather than generic simulator realism alone [8, 9, 30, 43]. We evaluate two prespecified settings: a mo-

Regime (sampling)	Baseline	Detector head (prespecified settings)	Doppler bands (Hz)
Labeled brain simulation scenarios (k-Wave; PRF 1500; 64-frame windows)	MC–SVD (energy fraction $e = 0.90$); adaptive SVD (SV similarity cutoff); block-wise local SVD (overlap-add; $e = 0.90$); RPCA; HOSVD (baselines scored by PD on residual)	tiles 8×8, stride 3; $L_t = 8$; robust covariance (Tyler-type) with diagonal loading $\lambda = 0.07$; matched-subspace flow-band score	$P_f = [30, 250]$, guard $[250, 400]$, $P_a = [400, 750]$
Gammex flow phantom (linear17; PRF 2500; $N = 17$ shots; pooled over cine frames)	baseline SVD band-pass keep $[2, 17]$; conventional short-ensemble comparators (PD/log-PD/Kasai)	tiles 8×8; $L_t = 16$; Tyler-type covariance with $\lambda = 0.07$; stride 6 (along) / 4 (across); matched-subspace flow-band score	$P_f = [150, 450]$, guard $[450, 700]$, $P_a = [700, 1200]$
ULM Zenodo 7883227 (PRF 1000; 128-frame blocks)	MC–SVD baseline with energy fraction $e = 0.975$ chosen by a one-time label-free sweep	tiles 8×8, stride 3; $L_t = 64$; SCM covariance with $\lambda = 0.07$; matched-subspace flow-band score	$P_f = [10, 150]$, guard $[150, 200]$, $P_a = [200, 500]$

Table 3: Prespecified operating settings for the main regimes. All settings are fixed per regime prior to reporting results; exact commands and exported metadata are provided in Supplementary Material.

bile setting and an intra-operative parenchymal setting. Both were calibrated against real-data operating characteristics and then fixed before held-out testing. A third, more surface-dominated intra-operative setting remains exploratory and is not used for quantitative claims. Here, "clinically anchored" means that nuisance profiles and operating settings are constrained by real-data fUS use cases and observed operating characteristics; it does not mean that the benchmark substitutes for prospective clinical validation.

For transparency, this benchmark is defined by explicit held-out splits and explicit label masks rather than by post hoc example selection. The structural checkpoint uses development seeds 125–126 to choose the final method families and settings, and held-out evaluation seeds 127–128 for the results in Table 4. The functional checkpoint uses a disjoint development split (seeds 221–222) and held-out evaluation split (seeds 223–224) under a common downstream readout. Structurally, each evaluated case exports the masks `mask_h1_pf_main`, `mask_h0_bg`, `mask_h0_nuisance_pa`, and `mask_h1_alias_qc`; the evaluation script computes the summary metrics directly from those masks under a fixed right-tail score convention.

For the structural task, the positive class is unaliased parenchymal Pf-band flow and the negative classes are (i) ordinary background and (ii) explicit nuisance structures such as superficial high-velocity or specular components. The three summary metrics in Table 4 are therefore plain-language clinical surrogates: $AUC_{\text{main/bg}}$ measures separation of target flow from ordinary background, $AUC_{\text{main/nuis}}$ measures separation of target flow from clinically relevant nuisance, and FPR_{nuis} @ $TPR_{\text{main}} = 0.5$ asks how many nuisance detections survive when the detector is tuned to retain half of the target positives. This last quantity is the most direct nuisance-tail metric in the table.

For the functional task, we use the same prespecified settings and a common background-controlled GLM-style readout on simulated task/null runs. Two comparisons are reported because they answer different questions: end-to-end pipeline ranking asks which full processing chain performs best after all residualizer-setting interactions, whereas the same-residual audit asks whether STAP improves the *same* residual cube under the same downstream readout.

Table 4 summarizes the prespecified held-out structural benchmark. The first two metric columns measure separability, while the last metric column directly measures nuisance leakage at a matched

Processing chain	Total runtime (s)	$\mathrm{AUC}_{\mathrm{main/bg}}$	$\mathrm{AUC}_{\mathrm{main/nuis}}$	$\mathrm{FPR}_{\mathrm{nuis}}$ at $\mathrm{TPR}_{\mathrm{main}} = 0.5$
HOSVD → Kasai	0.167	0.753	0.561	0.453
Adaptive Global SVD → matched-subspace default	0.638	0.851	0.953	0.000
RPCA → Kasai	1.783	0.837	0.442	0.593
RPCA → matched-subspace default	2.407	0.994	1.000	0.000

Table 4: Prespecified held-out SIMUS/PyMUST structural benchmark on held-out evaluation seeds under the prespecified evaluation setting. We highlight two conventional references and the two strongest fixed-head detector-family chains. $AUC_{\mathrm{main/bg}}$ is the area under the ROC curve for separating main signal from background. $AUC_{\mathrm{main/nuis}}$ is the area under the ROC curve for separating main signal from nuisance-labeled pixels. FPR_{nuis} @ $TPR_{\mathrm{main}} = 0.5$ is the nuisance false-positive rate at the threshold where main-signal true-positive rate equals 0.5. The key reader-facing pattern is that the two rows sharing the same downstream matched-subspace head both drive FPR_{nuis} @ $TPR_{\mathrm{main}} = 0.5$ to 0.000 despite using different upstream residualizers. On this benchmark, the simpler band-limited matched-subspace detector is therefore the fixed default; the adaptive and fully whitened modes are summarized later in the paper as separate members of the detector family.

main-signal sensitivity operating point. It is the main structural result for translational readers because it shows the tradeoff most likely to matter in practice: how much nuisance suppression is gained by moving from a conventional short-ensemble Doppler score to the current fixed band-limited matched-subspace head, and what runtime cost that gain carries on a workstation GPU. The most important pattern in the table is that both "Adaptive Global SVD → matched-subspace default" and "RPCA → matched-subspace default" drive the nuisance-leakage column to 0.000, even though the upstream residualizers are different. That common result is why we interpret the matched-subspace head itself, rather than only the choice of residualizer, as the main driver of the nuisance-tail gain on this benchmark. This benchmark is therefore the primary translational reference point in the paper: it shows that the simpler band-limited matched-subspace head is the strongest fixed default on held-out structural data, while the adaptive and fully whitened variants must be justified by additional value in clutter-heavier regimes.

For translational readers, the key question is not whether the method makes vessels look better on average, but whether fewer nuisance detections survive when the detector is tuned to preserve clinically relevant parenchymal flow. That is the practical meaning of FPR_{nuis} @ $TPR_{\mathrm{main}} = 0.5$: it asks how much structured false signal remains at a matched target-flow sensitivity operating point. Because fUS is already being used in intra-operative, bedside, cranial-window, and emerging mobile settings [1, 8, 20–22], that nuisance-leakage tradeoff is tied directly to usability and reader trust rather than to image aesthetics alone.

That per-window operating-point protocol is still an evaluation protocol, not a scanner-time thresholding rule. To address the more practical question directly, we also ran a fixed-calibration transfer check in which the threshold is learned once from separate calibration seeds 125–126 and then applied unchanged to the held-out evaluation seeds 127–128. Table 5 reports that result at nominal $\alpha = 10^{-3}$. On the mobile setting, the fixed matched-subspace head preserves nonzero held-out main recall (0.004 [0.002, 0.006]) while lowering realized nuisance FPR from 0.016 [0.015, 0.018] for PD to 0.003 [0.000, 0.006]. On the intra-operative parenchymal setting, both methods remain near the nominal background target, but only the fixed head suppresses nuisance leakage to 0.000 [0.000, 0.000], whereas PD remains at 0.489 [0.464, 0.514]. This is not the full prospective

Method	Mobile			Intra-operative parenchymal		
	$\mathrm{TPR}_{\mathrm{main}}$	realized $\mathrm{FPR}_{\mathrm{bg}}$	realized $\mathrm{FPR}_{\mathrm{nuis}}$	$\mathrm{TPR}_{\mathrm{main}}$	realized $\mathrm{FPR}_{\mathrm{bg}}$	realized $\mathrm{FPR}_{\mathrm{nuis}}$
Power Doppler on MC–SVD residual	0.000 [0.000,0.000]	0.002 [0.000,0.003]	0.016 [0.015,0.018]	0.012 [0.006,0.017]	0.001 [0.001,0.001]	0.489 [0.464,0.514]
Fixed matched-subspace head	0.004 [0.002,0.006]	0.002 [0.001,0.003]	0.003 [0.000,0.006]	0.006 [0.004,0.008]	0.001 [0.001,0.001]	0.000 [0.000,0.000]

Table 5: Fixed-calibration transfer on the prespecified held-out SIMUS/PyMUST structural benchmark at nominal $\alpha = 10^{-3}$. For each setting, a single threshold is learned once from pooled background negatives on separate calibration seeds 125–126 and then applied unchanged to the held-out evaluation seeds 127–128. Entries report mean [min,max] across the two held-out seeds, so this table is a direct non-retrospective threshold-transfer check rather than the per-window oracle calibration used for ROC reporting elsewhere.

scanner workflow yet, but it does show that the main structural default survives a separate-bank threshold transfer on the benchmark that anchors the paper.

To preempt the opposite over-reading of Table 4, we also ran a supplementary held-out SIMUS stress confirmation on the same mobile scene family using two perturbations that are closer to the nuisance mechanisms emphasized in the clinical fUS literature: cardiac-like low-frequency tissue pulsation and progressively shorter ensembles. The pulsation rows use a fixed 1.30 Hz cardiac-like nuisance term whose amplitude is increased over reference, moderate, and hard settings; this is intended to probe the same low-frequency tissue-motion regime emphasized by Demené *et al.* as critical for SVD clutter filtering and by Baranger *et al.* in neonatal/open-skull Doppler settings [11, 12]. The short-ensemble rows reduce the support from 64 frames to 48 and 32 frames to mimic clinically constrained acquisitions in which shorter windows are used to limit motion corruption or workflow burden [1, 22, 30]. For tractability, this confirmation fixes attention to the residualizer families that dominated the preliminary reduced-grid search (RPCA and adaptive-global SVD) and then reruns the held-out evaluation at the full reporting tier. Those stressors do not merely make the task numerically harder; they produce a clearer detector-family advantage than the earlier generic motion/reflectivity stress audit. On the held-out cardiac-pulsation confirmation, the strongest public comparator on those same stressed rows is RPCA→PD, with $AUC_{\mathrm{main/nuis}}$ 0.389 and FPR_{nuis} @ $TPR_{\mathrm{main}} = 0.5$ 0.673, whereas the strongest detector-family stack reaches 0.880 and 0.033. On the held-out short-ensemble confirmation, again relative to the strongest public comparator on the same stressed rows, the corresponding values improve from 0.399 to 0.812 and from 0.705 to 0.112. The row-wise pattern is also informative: the simpler matched-subspace default already wins the cardiac reference row and the shortest-ensemble row, whereas the fully whitened specialist takes the more aggressive pulsation rows and the intermediate short-ensemble row. We keep those rows in the supplement because they are stress-frontier evidence rather than a replacement for the prespecified main benchmark, but they make the practical point more cleanly: when the acquisition is pushed toward pulsatile nuisance and shorter support, the detector family retains a meaningful nuisance-control advantage over the strongest public baseline stacks evaluated under those same stress conditions.

Functional same-residual detector-head comparison. To test whether the fully whitened variant adds value beyond upstream residualization in function-priority settings, we report a compact same-residual comparison on the prespecified functional benchmark settings. For each fixed residual stream, we compare the fully whitened detector to conventional Doppler-style score heads using the same held-out data and the same downstream GLM-style readout.

Within the detector family itself, the robustly whitened Huber variant remains worth keeping

Benchmark setting	Best end-to-end chain	Same-residual score comparison	Interpretation
Intra-operative parenchymal	RPCA → PD-after-STAP	Improved in all 6 residual families	STAP helps both in the strongest end-to-end chain and in the same-residual comparison
Mobile	HOSVD → PD	Improved in 5 of 6 residual families	The best full pipeline depends on the acquisition setting, but STAP still improves most same-residual comparisons

Table 6: Same-residual functional detector-head comparison on prespecified benchmark settings under a common background-controlled GLM readout. "Best end-to-end chain" includes interactions between the upstream filter and the downstream score. "Same-residual score comparison" keeps the residual cube fixed and asks whether the fully whitened STAP-inspired detector improves the same downstream readout. This isolates detector-head behavior from upstream clutter suppression. Across these two prespecified settings, the held-out audit favors STAP in 11 of 12 family/setting comparisons, with the remaining case an effective tie.

even though it is not the structural default. On the held-out same-residual functional audit, it improves 5 of 6 mobile residual families for the direct STAP readout and 4 of 6 intra-operative residual families for the PD-after-STAP readout, with the remaining comparisons near-neutral rather than catastrophic. This is one prespecified whitening recipe carried across the functional, phantom, and labeled-stress-test regimes rather than a different score retuned for each dataset. That cross-regime consistency is the main reason we retain it as a distinct fully whitened variant inside the STAP-inspired detector framework. The adaptive variant described later sits between this fully whitened head and the simple fixed default: it does not replace the functional fully whitened result, but it provides a cleaner translational compromise than forcing full whitening everywhere.

Scope and low-FPR operating protocol. Quantitative detection performance is reported on labeled brain simulations, structurally labeled phantom regimes (Gammex flow phantom sequences; labels derived from B-mode geometry), and the prespecified held-out SIMUS/PyMUST benchmark using the strict low-FPR operating protocol of Section 4.1. Open real-IQ datasets are used primarily for bounded score-audit and motion-robustness evidence rather than for task ROC; the ULM section additionally includes a bounded localization-derived structural audit on real brain IQ, which we treat as an in-vivo vascular-fidelity check rather than as a decisive task-efficacy benchmark. Conditional execution is a decision-rule variant that uses a flow-proxy mask; leakage ablations and additional diagnostics are reported in Supplementary Material.

5.2 Controlled detector stress tests in labeled brain simulations

This subsection is intentionally supportive rather than primary. The SIMUS/PyMUST benchmark above is the main translational result in the paper. Here we turn to simpler labeled brain simulations only to isolate detector behavior under controlled nuisance stress tests. We report labeled low-FPR operating points on these scenarios using flow-evidence scores under a right-tail convention

(higher = more flow-like; thresholds calibrated on the high-score tail): a baseline power Doppler score and the proposed matched-subspace score (Notation). Baseline methods compare clutter suppression strategies (MC–SVD; adaptive SVD with a spatial singular-vector similarity cutoff; block-wise local SVD with overlap-add; RPCA; HOSVD) and are scored using the same power Doppler statistic; the matched-subspace detector family is evaluated on top of the MC–SVD residual. Across these scenarios, the PD-style baselines degrade sharply in the ultra-low-FPR tail, while the localized band-limited matched-subspace score retains measurable sensitivity at ultra-low FPR (i.e., $\mathrm{TPR}(\alpha) > 0$ at $\alpha = 10^{-4}$–10^{-3}), including in the structured-clutter setting where nuisance deliberately leaks into the positive class. Because these tables are meant to explain mechanism rather than headline performance, the main text keeps only the two more clinically interpretable simulation settings: an open-skull surface-artifact stress test and a structured-clutter leakage stress test. A third alias-dominant positive control showed the same qualitative trends but is omitted here for readability. Tables 7–9 should therefore be read as component audits on labeled stress tests, not as the primary translational result. Importantly, the MC–SVD+PD baseline row and the STAP row share the identical MC–SVD residual cube and differ only in the downstream score (PD versus matched-subspace), isolating the effect of scoring rather than the clutter-suppression stage. This is why the detector-component and detector-swap tables belong here: they explain *why* the main SIMUS/PyMUST benchmark later favors certain processing chains, but they do not replace that benchmark. A supplementary threshold-transfer audit on the labeled brain simulation stress tests still shows that the robustly whitened variant remains close to target in both the open-skull and structured-clutter settings. Those simulation-only checks are now secondary to the stronger deployment-transfer results reported in the main text on the held-out SIMUS/PyMUST benchmark and on the PALA-backed ULM audit.

Explicit limitations from the labeled-brain audits. Two points matter here. First, full whitening is not the structural default: on the prespecified held-out structural benchmark, the simpler band-limited matched-subspace detector remains stronger on nuisance-control metrics. Second, the robustly whitened variant is stable rather than brittle on these labeled-brain stress tests: open-skull realized FPR stays near target under both pairwise cross-window transfer (median 1.10×10^{-4}, $q_{75} = 1.64 \times 10^{-4}$ at $\alpha = 10^{-4}$) and a leave-one-window-out fixed calibration bank (median 1.20×10^{-4}), and the operating-point results are numerically unchanged across covariance-training trim $q \in \{0.05, 0.08, 0.10\}$. The remaining limitation is therefore about where whitening earns its extra complexity, not about broad instability of the fully whitened recipe itself.

Baseline fairness and prespecified settings. All baseline hyperparameters are fixed prior to reporting results. For the labeled brain simulation scenarios, the MC–SVD energy-fraction cutoff is selected once and then held at $e = 0.90$ for all reported windows. The comparator families span global low-rank filtering (MC–SVD and adaptive-global SVD), block-wise local filtering (fixed-energy and adaptive-local SVD), RPCA, and HOSVD. Baseline rows are then evaluated with the same right-tail Doppler score so that differences reflect the residualizer rather than a change in the downstream scoring rule. We do *not* retune these baseline families on the held-out windows to maximize their own strict-tail TPR after seeing the test cases; the main SIMUS/PyMUST benchmark instead gives each family a development-split configuration-selection step before held-out reporting, while the labeled brain stress tests use fixed representative settings chosen once and then frozen. The compact family summary is provided in Supplementary Material; in the main text

Method	Open-skull surface-artifact stress test			Structured-clutter leakage stress test		
	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$
PD on MC–SVD residual	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
Whitened power (no band)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
Unwhitened ratio (no whitening)	0.8689 (0.8674,0.8689)	0.8947 (0.8831,0.8985)	0.9389 (0.9166,0.9400)	0.9104 (0.8900,0.9189)	0.9373 (0.9058,0.9389)	0.9619 (0.9512,0.9689)
STAP (whitened ratio; pre-KA)	0.0730 (0.0008,0.0865)	0.1288 (0.0012,0.2222)	0.2218 (0.0015,0.4368)	0.5871 (0.5006,0.6682)	0.7205 (0.6048,0.7785)	0.8731 (0.8189,0.8758)

Table 7: Detector ablations on labeled brain simulation stress tests (MC–SVD residual, prespecified settings). All rows share the identical MC–SVD residual cube and differ only in the detector statistic: baseline PD, total power after tile-local whitening (no Doppler band partition), matched-subspace ratio without whitening (R=I), and the full STAP whitened matched-subspace ratio. The two column groups correspond to an open-skull surface-artifact stress test and a structured-clutter leakage stress test. Numbers are medians (IQR) over five disjoint 64-frame windows (offsets 0/64/128/192/256).

we keep the emphasis on what the comparisons mean scientifically rather than on implementation taxonomy.

Detector-component ablation (whitening versus band selectivity). To address the concern that the proposed score is simply "local adaptive whitening + band energy" under a new name, we report two detector ablations on the identical MC–SVD residual cube: (i) *whitened total power* (apply $R_i^{-1/2}$ but do not partition the Doppler spectrum) and (ii) an *unwhitened* matched-subspace ratio (set $R_i = I$ but retain the fixed flow-band subspace). Table 7 shows that whitening alone without band partition does not control the strict tail, while matched-subspace (banded) scoring is the dominant contributor in these labeled brain simulation scenarios under the prespecified settings. In the labeled brain simulations, the dominant gain appears to come from band-selective matched-subspace scoring itself, while adaptive whitening is helpful in some regimes but not uniformly beneficial. We therefore interpret the present method as a detector family whose main innovation is nuisance-aware score construction; local covariance adaptation is one important component of that family, but not the only source of gain.

Detector-swap composability audit (matched-subspace scoring on competing residuals). To isolate detector effects from the clutter-suppression stage, we also compute the matched-subspace STAP score on residuals produced by competing clutter filters (RPCA and HOSVD) and compare it to power Doppler scoring on the same residual under the identical window/mask protocol. Table 8 reports this paired-residual audit. Table 8 should not be read as contradicting Table 4. The labeled simulation detector-swap audit isolates detector-head behavior on the controlled k-Wave stress tests under fixed residual streams, whereas Table 4 reports end-to-end performance on the prespecified held-out structural benchmark. These answer different questions: detector-head fairness versus end-to-end deployment tradeoff. Under the prespecified STAP settings used throughout, strict-tail improvements do not automatically persist when composing STAP with more aggressive low-rank baselines. We therefore treat STAP as a second-stage detector defined for a coarse baseline residual (MC–SVD in the main brain-simulation pipeline), and use the controlled detector swap on the same MC–SVD residual as the primary isolation of detector effects.

Robust-whitening stability audit (labeled brain simulations). Because the whitened detector relies on tile-local covariance estimation, a natural concern is sensitivity to contaminated

Method	Open-skull surface-artifact stress test			Structured-clutter leakage stress test		
	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$
PD on RPCA residual	0.0000 (0.0000,0.0046)	0.0015 (0.0000,0.0088)	0.0042 (0.0042,0.0173)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
STAP on RPCA residual	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
PD on HOSVD residual	0.0000 (0.0000,0.0015)	0.0015 (0.0015,0.0015)	0.0046 (0.0046,0.0077)	0.0000 (0.0000,0.0015)	0.0000 (0.0000,0.0015)	0.0015 (0.0000,0.0031)
STAP on HOSVD residual	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0004)	0.0019 (0.0004,0.0019)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)

Table 8: Detector-swap fairness check on labeled brain simulation stress tests: PD vs matched-subspace STAP detector scores computed on the *same* residual produced by a competing clutter filter (paired bundles). The two column groups correspond to an open-skull surface-artifact stress test and a structured-clutter leakage stress test. Numbers are medians (IQR) over five disjoint 64-frame windows (offsets 0/64/128/192/256).

Trim setting	Open-skull stress test			Structured-clutter stress test		
	Within TPR@10^{-4}	Cross FPR med. (q75)	Fixed-cal FPR med. (q75)	Within TPR@10^{-4}	Cross FPR med. (q75)	Fixed-cal FPR med. (q75)
$q = 0.05$	0.852	1.10e-04 (1.64e-04)	1.20e-04 (1.39e-04)	0.917	9.96e-05 (2.44e-04)	7.97e-05 (1.99e-04)
$q = 0.08$	0.852	1.10e-04 (1.64e-04)	1.20e-04 (1.39e-04)	0.917	9.96e-05 (2.44e-04)	7.97e-05 (1.99e-04)
$q = 0.10$	0.852	1.10e-04 (1.64e-04)	1.20e-04 (1.39e-04)	0.917	9.96e-05 (2.44e-04)	7.97e-05 (1.99e-04)

Table 9: Robust-whitening stability audit for the prespecified Huber-whitened STAP variant on the labeled brain stress tests. The only changing parameter is the high-energy covariance-training trim q. Entries report within-window TPR at the strictest operating point ($\alpha = 10^{-4}$) together with realized FPR under pairwise cross-window transfer and a leave-one-window-out fixed calibration bank. The reported values are numerically unchanged across $q \in \{0.05, 0.08, 0.10\}$, indicating that the finalized advanced recipe is stable to modest covariance-training trim variation in these stress tests.

training support and threshold calibration. Table 9 examines those concerns directly for the prespecified Huber-whitened recipe by varying only the high-energy covariance-training trim over $q \in \{0.05, 0.08, 0.10\}$ and reporting both within-window performance and transfer behavior at the strictest operating point. The result is intentionally simple: the labeled-brain operating-point results are numerically unchanged across those trim settings, and open-skull realized FPR remains close to target under both pairwise cross-window transfer and a fixed calibration bank. We therefore treat contamination-robust covariance training as part of the finalized whitened recipe rather than as an unresolved instability of the method family. We further report a 1D sensitivity sweep over covariance diagonal loading λ, tile geometry, and slow-time aperture L_t on the canonical open-skull pilot (Supplementary Material).

Relaxed operating-point sanity anchor (labeled brain simulations). Several baseline rows in the supplementary labeled-brain strict-tail table take the value $\mathrm{TPR}(\alpha) \approx 0$ at $\alpha \leq 10^{-3}$. To keep the main Results focused on the main SIMUS/PyMUST benchmark and the strongest labeled stress tests, we place the relaxed-operating-point sanity table ($\alpha \in \{10^{-3}, 10^{-2}, 10^{-1}\}$) in the Supplementary Material and retain here only the representative tail-collapse visualization that explains why those strict-tail zeros occur.

Why PD-after-filter baselines can collapse in the strict tail. A baseline clutter filter can yield nontrivial separation at moderate operating points yet still fail at ultra-low FPR if the background score distribution has an extreme right tail: the background-calibrated threshold τ_α (e.g. the 99.9th percentile of H_0 at $\alpha = 10^{-3}$) can lie above essentially all H_1 pixels, yielding $\mathrm{TPR}(\alpha) \approx 0$ by construction. For example, in a representative open-skull labeled brain window

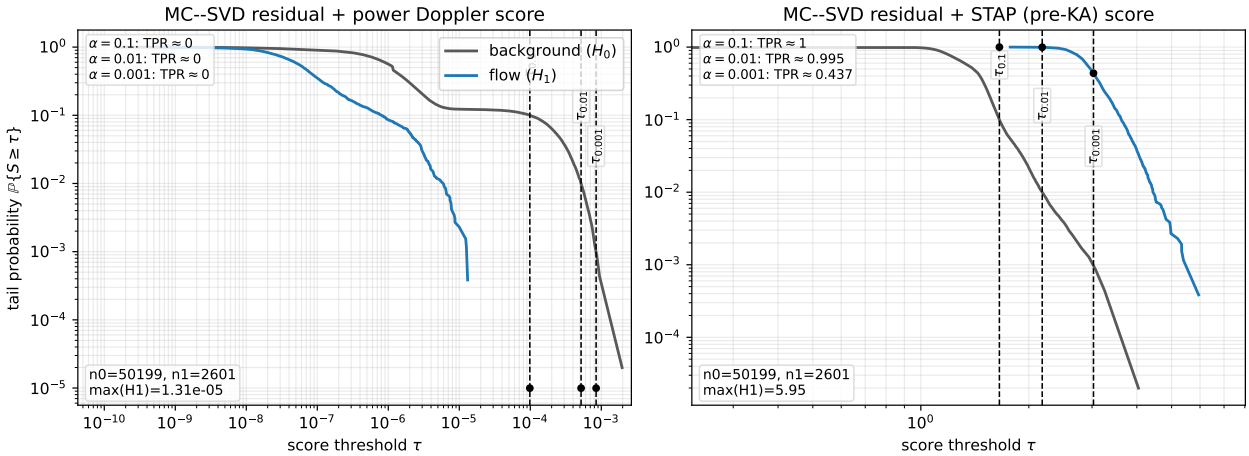

Figure 7: Representative open-skull labeled brain window illustrating strict-tail collapse of PD-after-filter baselines. Curves show empirical tail probabilities $\mathbb{P}\{S \geq \tau\}$ for background (H_0) and flow (H_1) score distributions versus score threshold τ, for (left) the baseline power Doppler score on the MC–SVD residual and (right) the pre-KA matched-subspace STAP score computed on the same residual. Vertical lines mark background-calibrated thresholds τ_α at $\alpha \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ and markers show the corresponding $\mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}$. For the baseline PD score, $\tau_{10^{-3}}$ lies above $\max(H_1)$, yielding TPR ≈ 0 by construction; the STAP score retains nonzero sensitivity at the same strict-tail operating point.

the MC–SVD+PD baseline background quantiles are $q_{0.99} \approx 5.9$ and $q_{0.999} \approx 9.3$ while the flow $q_{0.999} \approx 0.094$, so no flow pixels exceed the $\alpha = 10^{-3}$ threshold; Figure 7 visualizes the corresponding tail geometry ($\tau_{10^{-3}} > \max(H_1)$) for the baseline PD score and contrasts it with the STAP score computed on the same residual. This tail-dominance failure mode motivates both the strict low-FPR operating discipline and detector heads that explicitly reshape the tail, beginning with band-limited matched-subspace scoring and, when needed, tile-local covariance whitening.

Latency and conditional execution. We also report representative runtimes for the final detector family evaluated in this manuscript rather than for an older whitening-heavy research configuration. In this evaluation, the labeled-brain stress tests use the simple unwhitened matched-subspace default with conditional execution, the Shin real-IQ evaluation uses the adaptive variant, and Gammex uses the short-ensemble raw-baseline plus unwhitened matched-subspace head. Conditional execution still changes the decision rule by deciding which tiles receive the adaptive detector; we therefore treat it as part of the reported detector configuration rather than as a cost-free acceleration of the same method. Unless otherwise stated, any conditional-execution reporting uses the same leakage-controlled, disjoint-window proxy discipline as the detection results.
Latency is measured by replaying multiple windows or frames in a single GPU process and reporting *steady-state* means over windows/frames 2..N. The first replay window/frame is excluded because it includes one-time startup overheads. We time the baseline and detector stages separately (excluding disk I/O), and "Total" reports baseline+detector wall time. "Budget" denotes the acquired slow-time duration of the corresponding ensemble: $64/1500 \approx 42.7$ ms for the brain windows, $128/1000 = 128$ ms for Shin, and $17/2500 = 6.8$ ms for the Gammex cines. The reported real-time factor (RTF)

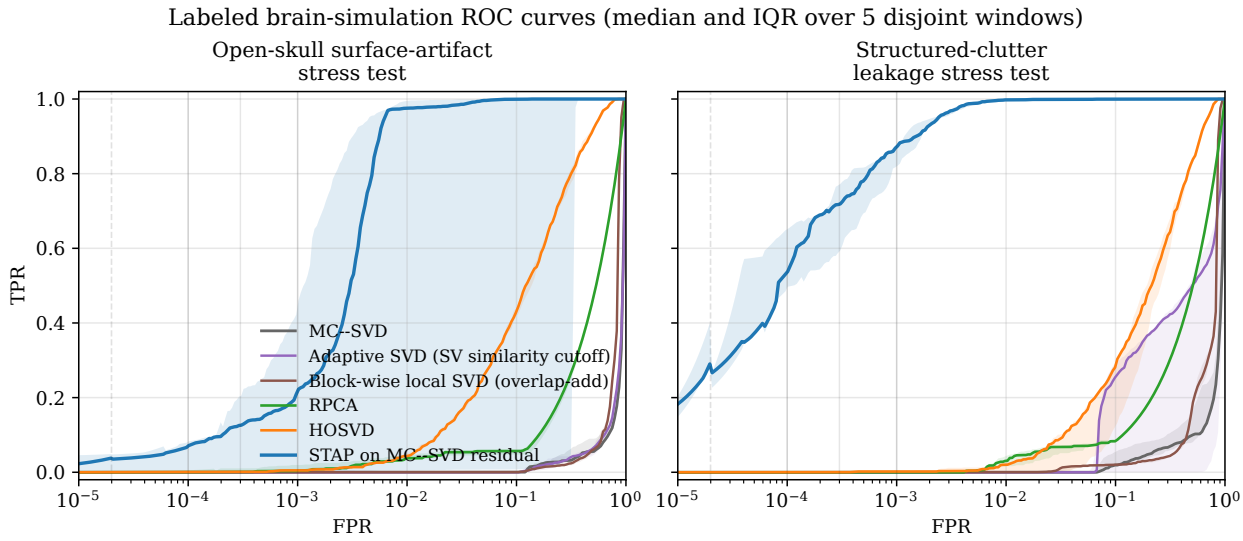

Figure 8: Controlled labeled-brain stress tests: ROC-style curves (TPR vs FPR on a log-x axis) under the same 64-frame window protocol used for the low-FPR operating-point reporting in the main text. Curves include MC–SVD, adaptive and local SVD baselines, RPCA, HOSVD, and STAP; each curve shows the median and interquartile range over five disjoint windows. For readability the main figure keeps the two more clinically interpretable simulation settings (open-skull surface artifact and structured-clutter leakage); the alias-dominant positive control is reported in the Supplementary Material. The moderate-FPR region illustrates that several PD-after-filter baselines can show separation at relaxed operating points while saturating at strict-tail operating points. The vertical dashed line indicates the supported floor $fpr_{\min} = 1/n_{\text{neg}}$.

is therefore Total/Budget.

Portable-workstation feasibility (tempered). A common critique of covariance-adaptive fUS scoring is that local covariance conditioning and the associated matrix factorizations are prohibitively expensive when applied densely across an image grid. Table 10 shows that, once the detector family is timed in the final reported configuration rather than in a whitening-heavy research setup, the picture is substantially better on an RTX 4080 SUPER. All three labeled-brain stress tests run below the full acquisition-PRF budget (RTF $\approx$ 0.86–0.88), the Shin rat-brain real-IQ evaluation also remains below budget (RTF $\approx$ 0.84), and both Gammex views now run below the 17-shot cine-frame budget (RTF $\approx$ 0.88 along-view and $\approx$ 0.64 across-view).[1] These results suggest that nuisance-aware matched-subspace scoring need not require server-class hardware, while still making clear that throughput depends on the chosen detector variant and should be interpreted with the explicit decision-rule caveats of conditional execution and adaptive whitening.
A leakage ablation confirms that conditional execution can inflate apparent performance under same-window proxy gating, while leakage-controlled gating yields a small TPR decrease (order 10^{-3}) at the target FPRs. We therefore treat conditional execution as a distinct decision rule

[1]Latency is reported from steady-state replay (windows/frames 2..N) and excludes startup overhead and disk I/O. Table 10 reports the final detector family used in the manuscript rather than a single worst-case fully whitened configuration applied everywhere.

Scenario	Baseline (s)	Detector (s)	Total (s)	Budget (s)	RTF
Open-skull surface-artifact stress test (64-frame window)	0.024	0.012	0.037	0.043	0.86
Alias-dominant positive control (64-frame window)	0.025	0.012	0.038	0.043	0.88
Structured-clutter leakage stress test (64-frame window)	0.026	0.012	0.037	0.043	0.87
Shin rat-brain real-IQ evaluation (128-frame window)	0.006	0.102	0.108	0.128	0.84
Gammex flow phantom, along view (17-shot cine frame)	0.000	0.006	0.006	0.007	0.88
Gammex flow phantom, across view (17-shot cine frame)	0.000	0.004	0.004	0.007	0.64

Table 10: Steady-state latency summary on an NVIDIA GeForce RTX 4080 SUPER 16GB for the final detector family evaluated in this manuscript. Times are means over windows/frames 2..N, exclude disk I/O, and compare baseline and detector stages under the reported detector configuration rather than under an earlier whitening-heavy timing setup.

rather than a cost-free speedup and place the full open-skull leakage table with bootstrap intervals in the Supplementary Material.

5.3 Phantom acquisitions (Gammex flow phantom): structural ROC and within-ensemble motion ladders

Dataset and acquisition parameters. We use the open-access Doppler Twinkling Artifact database [38], which provides complex beamformer output from phantom acquisitions on a clinical scanner. Two parts of that dataset matter here for different reasons. The *Gammex flow phantom* provides a simple vessel-like geometry that lets us derive independent structure-based flow/background labels from B-mode anatomy alone. The *artificial calculi* sequences provide an artifact-heavy setting in which twinkling is expected and are therefore useful for auditing whether the shrink-only KA layer behaves conservatively rather than spuriously inflating scores. Each cine frame contains a contemporaneous B-mode image and a short-ensemble color-flow (CFM) slow-time block with $N \in \{5, 9, 17\}$ pulses. Leonov *et al.* explicitly report PRF for some acquisition families, including the transverse linear-probe Gammex acquisition, but also note that PRF information is not consistently preserved across the database. We therefore fix PRF once per sequence family, using the source-paper value where it is explicitly documented and otherwise recording a sequence-level override in bundle metadata. Band-discretization sanity checks disable KA ($w \equiv 1$) when discretized Pf/Pa bands are empty or overlapping (Section 3). Decoding/QC details are provided in Supplementary Material (PRF provenance and decoder sanity); reproduction commands and manifests are provided in Supplementary Material.

B-mode-only structural tube masks (independent H_1/H_0). For the Gammex flow phantom sequences we derive evaluation masks from B-mode structure only (no Doppler/STAP outputs). Using acquisition geometry metadata that specifies the CFM region-of-interest (ROI) within the B-mode grid (beam and sample offsets and any beam subsampling), we map the B-mode ROI onto the CFM grid and segment the tube lumen on a median log-envelope reference image. This yields (i) a conservative lumen (flow) mask, eroded to avoid wall pixels and (ii) a background mask that excludes a guard band around the lumen and bright reflectors. Mask provenance and QC metrics are recorded with each experiment; Figure 9 shows representative overlays. These structural labels support a defensible phantom ROC.

Frozen short-ensemble Doppler profile and baseline fairness scores. For the linear-probe Gammex sequences used here ($N = 17$ shots; PRF values listed in Supplementary Material), we use a prespecified short-ensemble operating profile: baseline SVD band-pass keep range $[2, 17]$, STAP aperture $L_t = 16$, tiles 8×8, a Tyler-type robust covariance estimator, and diagonal loading $\lambda = 0.07$. To keep the profile auditable while avoiding sample-support limitations on small ROIs and excessive overlap/mixing on large ROIs, we use a deterministic geometry-only stride policy: choose the largest tile stride ≤ 6 such that the implied tile count is ≥ 500. On the two Gammex views this yields stride 6 (along; large ROI) and stride 4 (across; small ROI). Band definitions are set to $P_f = [150, 450]$ Hz and $P_a = [700, 1200]$ Hz. For baseline context on short ensembles, we also report conventional comparators derived from the baseline-filtered cube (power Doppler, log power Doppler, and Kasai lag-1 power). Reproducibility details are provided in Supplementary Material.

Structural-label low-FPR ROC (Gammex). At the strict FPR levels reported here, several conventional baselines collapse not because they are universally nonfunctional, but because residual nuisance structure dominates their upper score tail and exhausts the allowed false-positive budget before true flow targets are recovered. We therefore pool pixel scores across cine frames and report

low-FPR operating points using right-tail thresholding on pooled background pixels, then evaluate TPR on pooled lumen pixels. In these short-ensemble Doppler acquisitions, the background set includes strong twinkling/artifact pixels whose right-tail scores dominate conventional baselines; consequently, low-FPR thresholds at $\alpha \leq 10^{-3}$ can lie above essentially all lumen pixels, yielding numerically zero baseline TPR at the reported operating points. Figure 10 visualizes this effect for a representative baseline comparator. To avoid treating pooled pixels as independent samples, we report 95% confidence intervals via a frame-level bootstrap: we resample cine frames as the unit of variation while holding the pooled threshold fixed. Table 11 summarizes a broader same-residual score comparison for the two Gammex views under those same settings: power Doppler and Kasai on the baseline residual, plus three STAP-family score variants evaluated on that identical residual cube (whitened power, unwhitened matched-subspace ratio, and the full whitened matched-subspace ratio). For compactness we omit log-PD from the main table because it is monotone-equivalent to PD here and numerically identical at the reported operating points; the full pooled summaries remain available in Supplementary Material. The practical relevance is already visible at a conventional operating point. On the representative Figure 2 frame at matched background FPR 10^{-2}, the baseline thresholded map contains 884 suprathreshold pixels and none lie inside the structural lumen mask, whereas the STAP score recovers 586 lumen pixels while holding the same background allowance. The stricter operating points reported below should therefore be read as a cleaner stress test of the same map-trustworthiness problem rather than as a claim that only 10^{-4} matters in practice. This decomposition clarifies the phantom fairness story. First, the baseline filter is not removing lumen flow: once the downstream score is changed on the same residual cube, both whitened-power and the full STAP score retain nonzero sensitivity. Second, local whitening is the decisive ingredient in this artifact-heavy phantom: removing whitening collapses the detector strongly on both views, especially the along view. Third, the full flow-band matched-subspace ratio is *not* the strongest score on Gammex: total whitened slow-time power is stronger on both views under this short-ensemble setting, while the matched-subspace ratio is more conservative. We therefore treat Gammex as a structurally labeled artifact-tail stress test that supports the value of local adaptive whitening and same-residual score analysis, not as proof that the matched-subspace ratio is universally the best choice in every phantom regime. The view asymmetry still matters: the across view provides the clearest structural-label phantom evidence, whereas the along view remains a harsher artifact-tail stress test under this short-ensemble profile. To bound the remaining fairness objections, we also keep a small untuned sanity audit in Supplementary Material: on the first 20 cine frames, applying STAP directly to raw registered IQ changes TPR@10^{-3} only from 3.48×10^{-4} to 2.61×10^{-4} on the along view and from 0.202 to 0.205 on the across view; perturbing the frozen along-view PRF from 2000 to 3000 Hz leaves TPR@10^{-3} unchanged at 3.48×10^{-4} in that limited audit. This phantom section should therefore still be read as an artifact-tail stress test under a prespecified short-ensemble profile, not as proof that every conventional head is universally zero under any plausible PRF or residualizer.

A compact STAP-only ablation and PRF-override sensitivity check for Gammex are provided in the Supplementary Material. We keep them supplementary because they are profile-sanity diagnostics rather than the primary structural-label result.

Within-ensemble motion ladders (labeled, real acquisition). To obtain a labeled motion-robustness curve on a real acquisition without external motion ground truth, we inject synthetic *within-ensemble* motion directly into the CFM cube: for each cine frame we apply a per-shot

Method / score	Along (linear17 @ PRF 2500) TPR@$\alpha = 10^{-4}$	TPR@$\alpha = 3\times10^{-4}$	TPR@$\alpha = 10^{-3}$	Across (linear17 @ PRF 2500) TPR@$\alpha = 10^{-4}$	TPR@$\alpha = 3\times10^{-4}$	TPR@$\alpha = 10^{-3}$
Baseline (power Doppler)	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]
Baseline (Kasai lag-1 power)	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]
Detector ablation (whitened power)	0.0012 [0.0011,0.0014]	0.0119 [0.0114,0.0124]	0.0532 [0.0521,0.0543]	0.4820 [0.4776,0.4863]	0.5155 [0.5120,0.5191]	0.5813 [0.5784,0.5842]
Detector ablation (unwhitened ratio)	0.00000 [0.00000,0.00001]	0.00000 [0.00000,0.00001]	0.00001 [0.00000,0.00002]	0.0574 [0.0534,0.0616]	0.0602 [0.0564,0.0645]	0.0653 [0.0612,0.0693]
STAP (whitened matched-subspace)	0.00000 [0.00000,0.00000]	0.00003 [0.00000,0.00007]	0.00031 [0.00020,0.00042]	0.0491 [0.0470,0.0514]	0.0998 [0.0970,0.1026]	0.2237 [0.2197,0.2280]

Table 11: Gammex flow phantom structural-label operating points using B-mode-derived structural masks. All scores are right-tail (higher indicates more flow evidence). This is a same-residual detector-head comparison: every row is evaluated on the identical baseline SVD band-pass residual cube, changing only the detector head. The baseline rows are power Doppler and Kasai lag-1 power on that residual; the detector ablations replace the STAP matched-subspace score with either total whitened slow-time power (no Doppler band partition) or the same flow-band matched-subspace ratio without covariance whitening ($R = I$). Thresholds are chosen per method on pooled background pixels to match each target FPR and evaluated on pooled lumen pixels. Brackets show 95% frame-bootstrap CIs ($n = 2000$ resamples over cine frames) at the fixed pooled threshold. Pooled background counts are $n_{\text{bg}} = 3.85 \times 10^6$ for the along view ($fpr_{\min} \approx 2.6 \times 10^{-7}$) and $n_{\text{bg}} = 1.97 \times 10^6$ for the across view ($fpr_{\min} \approx 5.1 \times 10^{-7}$). Reproducibility details are provided in Supplementary Material.

Regime	KA active / inert bundles	median p_{shrink}	background-tail rate at pre-KA $\tau_{10^{-3}}$ (pre→post)	background-tail clusters at $\tau_{10^{-3}}$ (pre→post)
Twinkling calculi (PRF 500; $L_t = 8$)	50/0	0.102	$1.00 \times 10^{-3} \to 9.38 \times 10^{-4}$	888 → 832

Table 12: Compact quantitative KA audit on the artifact-heavy Twinkling calculi regime. Under the prespecified short-ensemble profile, the shrink-only prior is active on all 50 bundles (49 safety-only, 1 uplift-eligible), and the protected set is preserved exactly (maximum absolute score change = 0 on protected pixels). The fixed-threshold audit uses the pre-KA threshold $\tau_{10^{-3}}$ chosen so that the background proxy satisfies $\mathbb{P}\{S_{\text{stap,pre}} \geq \tau_{10^{-3}}\} = 10^{-3}$; at that same threshold, KA modestly compresses the background tail without score inflation. This is a quantitative hygiene audit, not a primary ROC claim. Extended region/reason panels and representative pre/post tail plots remain in Supplementary Material.

rigid translation $(\Delta y_t, \Delta x_t)$ to the N CFM shots prior to baseline and STAP, with motion kinds `rw` (random walk) and `step`. Because the structural tube masks are derived from B-mode and fixed per frame, this produces a labeled "motion ladder" where the only change is within-ensemble corruption. We generate motion-corrupted ensembles (85 cine frames; amplitudes 0–2 px) and aggregate using frame-level bootstraps for CIs together with an explicit amp=0 equivalence check against the non-motion reference. Figure 12 shows the resulting curves; reproducibility details are provided in Supplementary Material.

5.4 Compact quantitative KA audit on artifact-heavy calculi

Because the main labeled detection tables report the pre-KA STAP score, we include one compact quantitative KA audit in a regime where the prior is expected to act conservatively rather than to introduce a new headline efficacy claim. We use the artifact-heavy Twinkling calculi sequences under the prespecified short-ensemble setting and summarize activation, protected-set invariance, and fixed-threshold tail hygiene.

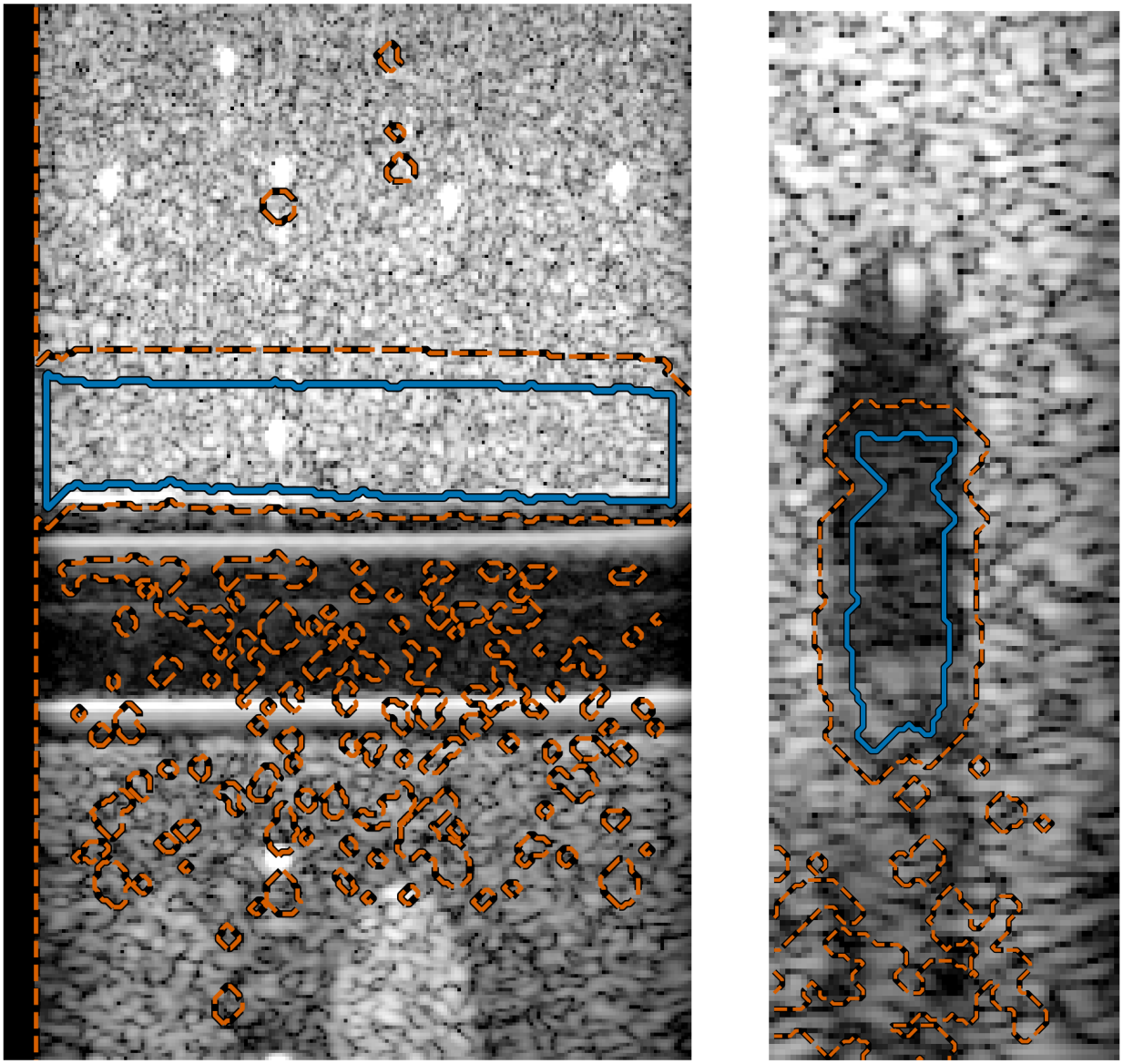

Figure 9: Gammex flow phantom structural evaluation masks derived from B-mode only. Left: along-linear view; right: across-linear view. The lumen mask is intentionally eroded to avoid wall/partial-volume pixels, and the background mask excludes a guard band around the lumen and bright reflectors. Images show the CFM-aligned B-mode ROI with contour overlays of the tube lumen (flow) mask (blue) and background mask (orange, dashed) used for structural-label ROC reporting. Reproducibility details are provided in Supplementary Material.

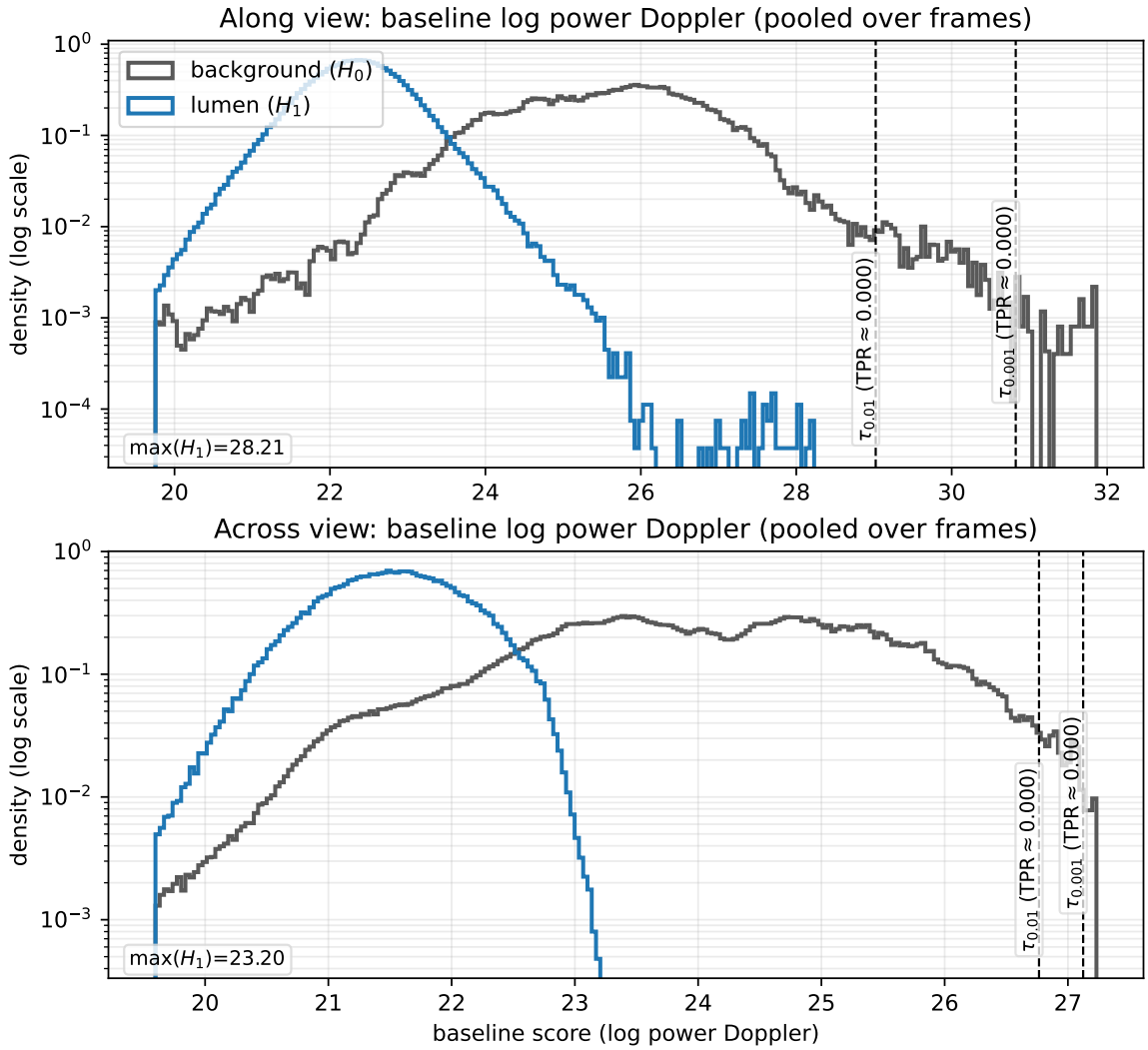

Figure 10: Why conventional baselines can have zero TPR at ultra-low FPR on Gammex short-ensemble data. The plot shows pooled score distributions for the baseline log power Doppler comparator on lumen (H_1; structural tube mask) versus background (H_0; structural background mask), for both along and across views. Vertical lines indicate background-calibrated right-tail thresholds τ_α at $\alpha \in \{10^{-2}, 10^{-3}\}$ (the main text reports $\alpha \leq 10^{-3}$; $\alpha = 10^{-2}$ is shown here for context). On both views the background tail threshold lies above $\max(H_1)$, so $\mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}$ is numerically zero at these tail operating points.

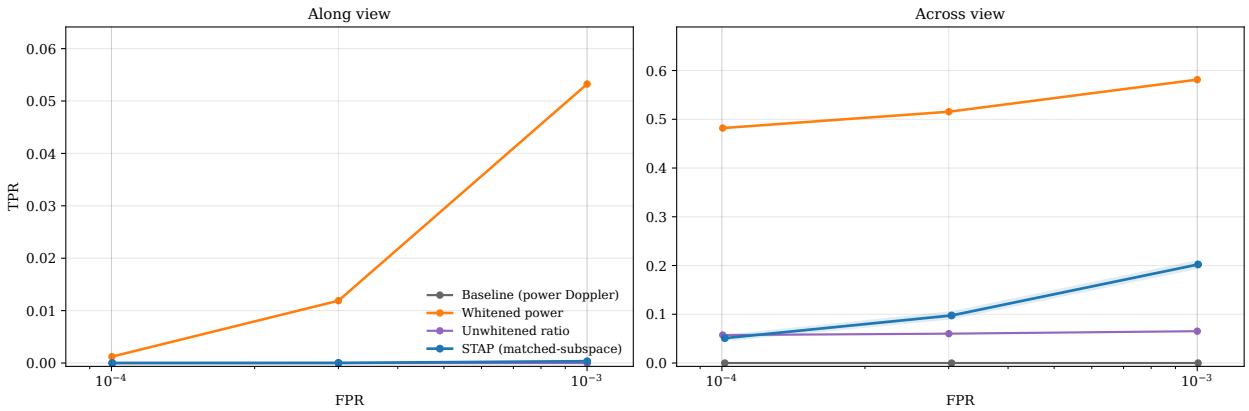

Figure 11: Gammex flow phantom same-residual score comparison in the strict low-FPR region used for the main phantom table. Curves connect the three reported operating points for baseline power Doppler, whitened power, unwhitened matched-subspace ratio, and the full pre-KA STAP matched-subspace score. The two panels use panel-specific TPR ranges so that the operating-point differences are readable.

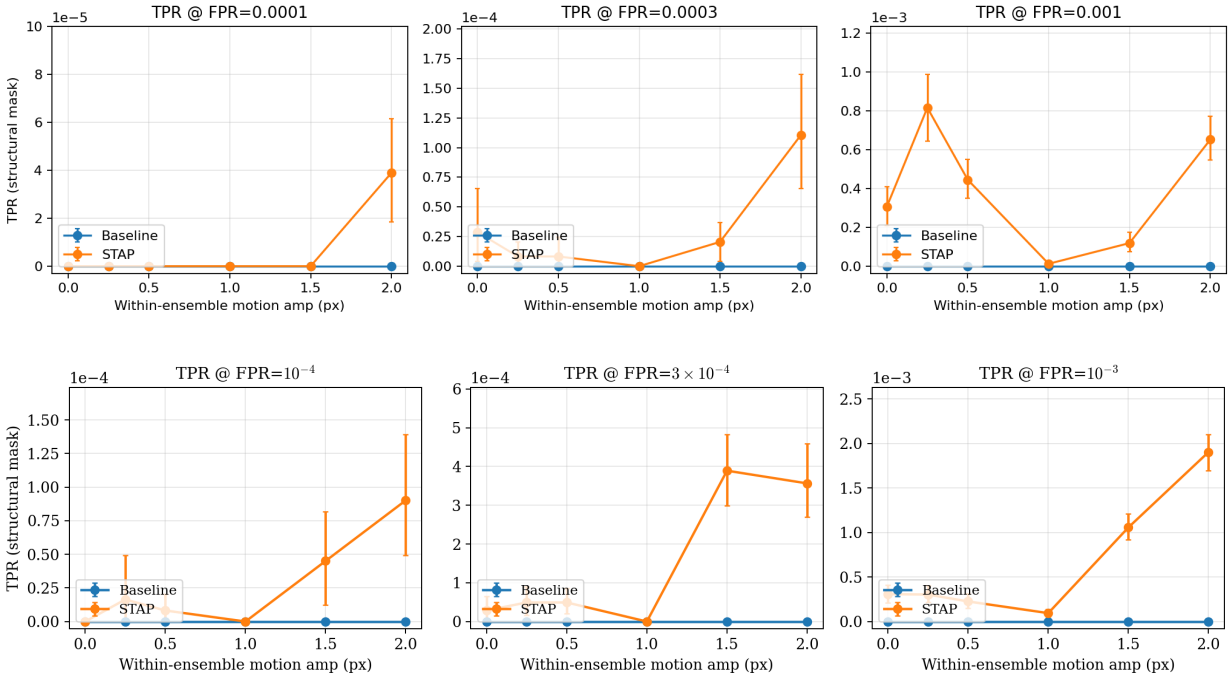

Figure 12: Gammex flow phantom within-ensemble motion ladder on the along-linear PRF=2500 sequence (85 frames). Left: random-walk per-shot translation; right: step translation. Each sub-panel reports TPR vs motion amplitude at a fixed FPR target using B-mode structural masks, with the y-axis zoomed to the observed TPR range so that small changes remain visible. Shaded bands are 25–75% bootstrap intervals over cine frames. Reproducibility details are provided in Supplementary Material.

5.5 ULM Zenodo 7883227 (real kHz IQ): baseline calibration and nonrigid motion robustness

Dataset and loader. We use a real kHz IQ dataset released for ultrasound localization microscopy benchmarking [33, 34] and distributed as HDF5 blocks with accompanying acquisition metadata. Each block contains a dataset `iq` of shape $(2, H, W, T) = (2, 256, 180, 800)$ float32 representing I/Q over $T = 800$ slow-time frames. We load blocks on demand and return complex IQ as a cube of shape (T, H, W) (reproduction details in Supplementary Material).

One-time label-free baseline selection. For ULM we select a single MC–SVD energy-fraction cutoff by a one-time label-free sweep on blocks 1–3 over frames 0:128, and then freeze it for all subsequent ULM motion experiments. The sweep uses baseline-only runs and relies on label-free integrity and stability criteria (no task labels); it yields a frozen energy fraction $e = 0.975$ (full sweep details in Supplementary Material).

Nonrigid motion injection and fairness alignment ablation. We evaluate motion robustness by injecting controlled motion directly into the beamformed IQ cube and re-running baseline+STAP under a frozen ULM profile (PRF = 1000 Hz; $L_t = 64$; tiles 8×8; stride 3; SCM covariance; diagonal loading $\lambda = 0.07$; baseline MC–SVD with frozen $e = 0.975$). Because motion is injected at the beamformed IQ stage, this test stresses post-beamforming robustness to residual motion mismatch; it does not model channel-domain decorrelation or skull-induced phase-aberration physics. Motion ladders span two nonrigid families: `brainlike` (rigid random-walk translation plus a smooth, depth-decaying elastic warp with temporally correlated coefficients and micro-jitter) and `elastic` (nonrigid-only; rigid and micro-jitter disabled). For each amplitude we compute label-free robustness endpoints on the baseline score S_{base} and the matched-subspace STAP score $S_{\text{stap,pre}}$ relative to the no-motion reference under three *common* alignment modes (none / common B-mode / injected GT proxy) plus a diagnostic per-method "self" alignment. We additionally report degeneracy checks (std ratios, top-k overlap, and tail quantile ratios) to rule out "stability by collapse." Figure 13 shows the main `brainlike` sweep; additional motion variants are provided in Supplementary Material.

Endpoint definition. For a score map $S_{\text{meth}}(a)$ at motion amplitude a and the corresponding no-motion reference $S_{\text{meth}}(0)$ (same block), we compute $\rho_{\text{meth}}(a)$ as the Pearson correlation between the aligned maps on a central crop Ω_a that excludes a margin of $\lceil a \rceil + 2$ pixels to avoid boundary artifacts from warping. Correlations are computed directly on the exported right-tail scores (no log transform) and use the same crop/alignment for baseline and STAP in the "common" alignment modes.

Representative quantitative endpoints (ULM). At a 3 px motion amplitude under the injected GT alignment proxy, the median correlation on the baseline score S_{base} drops to ≈ 0.20 (`brainlike`), while the STAP score $S_{\text{stap,pre}}$ remains ≈ 0.75 (medians over blocks 1–3; full tabular summaries and additional motion variants are provided in Supplementary Material). Degeneracy checks confirm that this stability is not explained by map collapse: the median std ratio and 0.999-quantile ratio of $S_{\text{stap,pre}}$ remain nontrivial across amplitudes, and top-k overlap decays gradually rather than saturating to trivial values.

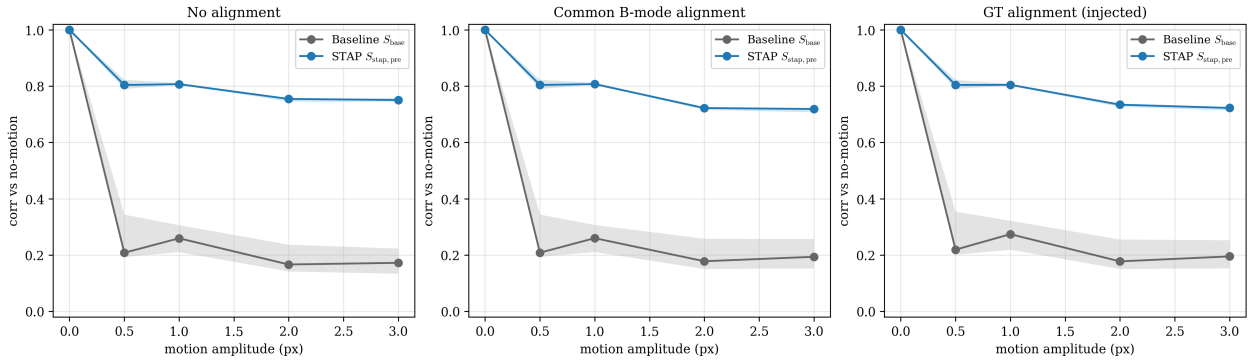

Figure 13: ULM 7883227 motion sweep (blocks 1–3; frames 0:128; frozen baseline $e = 0.975$): `brainlike` motion (rigid + elastic). Each panel reports correlation vs the no-motion reference under a different alignment mode (none / common B-mode / injected GT proxy) for the baseline score S_{base} and the STAP score $S_{\text{stap,pre}}$; shaded bands show median±IQR over blocks. Reproducibility details are provided in Supplementary Material.

Localization-derived structural-label audit on the same dataset. To obtain a more anatomical structural readout on the same open real-IQ data, we use the published PALA rat-brain ULM example as an external vascular reference derived from microbubble localization microscopy rather than from the Doppler scores being evaluated [33, 34]. We first reproduce the corresponding short-horizon PALA PowDop rendering from the Zenodo IQ blocks using the published odd-block selection and SVD cutoff, register that published PowDop crop back to the Zenodo IQ field of view, and then downsample the published super-resolution MatOut vascular rendering onto the short-window evaluation lattice. From that registered localization map we derive a conservative vessel-core versus perivascular-shell label pair: the vessel mask is a peak-based core mask, and the background mask is the surrounding shell, so the endpoint asks whether a short-window detector concentrates score on the vascular core rather than the immediately adjacent parenchyma. That one-time PALA-to-IQ registration is part of the label construction only. For the short-window detector itself we keep frame-to-frame registration disabled in the frozen audit, because re-enabling it changes the structural ROC only below the fourth decimal place while adding substantial latency.

Under this bounded but anatomically grounded endpoint, the strongest specialist score is no longer PD or Kasai. In the completed 10-block / 20-window audit with short 64-frame windows, the fully whitened matched-subspace specialist reaches AUC 0.526 [0.514, 0.536], compared with 0.503 [0.493, 0.513] for PD and 0.502 [0.492, 0.512] for Kasai. At matched 70% vessel-core recall, the same specialist slightly lowers perivascular-shell false positives from 0.701 [0.687, 0.715] for PD and 0.704 [0.689, 0.717] for Kasai to 0.690 [0.670, 0.710]. The effect is bounded and modest in absolute magnitude, but consistently positive across the audited windows. That detector choice is also scientifically coherent with the endpoint: this PALA-backed audit rewards concentration on the localization-derived vascular core relative to the immediately adjacent shell, whereas the held-out SIMUS nuisance-control benchmark rewards the strongest fixed downstream robustness against structured nuisance, so the winning head here is the fully whitened matched-subspace specialist rather than the fixed matched-subspace default. On an RTX 4080 SUPER, the same prespecified 64-frame no-registration specialist configuration runs at 41.3 ms steady-state total for a 64 ms acquisition budget, so this bounded in-vivo structural result is not tied to an offline-only

Score	AUC (H1 vs H0)	TPR@$\alpha = 10^{-1}$	TPR@$\alpha = 10^{-2}$	TPR@$\alpha = 10^{-3}$
Baseline (power Doppler)	0.503 [0.493,0.513]	0.105 [0.097,0.113]	0.009 [0.007,0.011]	0.000 [0.000,0.000]
Baseline (log-power Doppler)	0.503 [0.493,0.513]	0.105 [0.097,0.113]	0.009 [0.007,0.011]	0.000 [0.000,0.000]
Baseline (Kasai lag-1 magnitude)	0.502 [0.492,0.512]	0.106 [0.098,0.114]	0.008 [0.007,0.011]	0.000 [0.000,0.000]
Detector-filtered power Doppler	0.077 [0.073,0.081]	0.007 [0.004,0.012]	0.000 [0.000,0.000]	0.000 [0.000,0.000]
Matched-subspace detector (whitened specialist)	0.526 [0.514,0.536]	0.147 [0.141,0.152]	0.023 [0.018,0.027]	0.003 [0.001,0.007]

Table 13: Structural-label ROC on real in vivo rat-brain IQ from ULM Zenodo 7883227. For each evaluation block, vessel and background masks are derived once from an external super-resolved vascular reference derived by registering the published PALA rat-brain ULM example back onto the Zenodo IQ grid and downsampling the registered MatOut rendering to the short-window evaluation lattice; short 64-frame windows from the held-out block are then scored without additional frame-to-frame registration on the same MC–SVD residual cube using different downstream detector heads. Entries report window-level means with 95% bootstrap CIs over windows. Reproducibility details are provided in Supplementary Material.

Score	core TPR	realized shell FPR
Baseline (power Doppler)	0.000 [0.000,0.000]	0.001 [0.000,0.002]
Baseline (Kasai lag-1 magnitude)	0.000 [0.000,0.000]	0.001 [0.000,0.002]
Whitened matched-subspace specialist	0.002 [0.000,0.004]	0.001 [0.000,0.002]

Table 14: Fixed-calibration transfer on the PALA-backed ULM structural audit at nominal $\alpha = 10^{-3}$. A threshold is learned once from pooled perivascular-shell negatives on one 5-block calibration bank and then applied unchanged to the complementary 5 held-out blocks; the reported values pool the two complementary block splits and show window-level means with 95% bootstrap CIs over held-out windows. This is therefore a genuine non-retrospective threshold-transfer check on real in-vivo IQ rather than per-block retuning.

implementation. This remains a bounded in-vivo structural audit rather than a task-activation or hemodynamic-response benchmark: the labels come from a published localization-derived vascular rendering on the same acquisition, not from an independent modality or functional task.

Figure 1 gives the visual counterpart to this result by showing the localization-derived structural masks together with a representative same-window comparison between baseline power Doppler and the whitened matched-subspace specialist. Table 13 gives the formal held-out quantitative summary; the figure is intended only to show how the two maps differ anatomically on a representative window.

A second deployment-oriented check uses the same bounded structural endpoint but removes per-window retuning. We split the 10 audited blocks into two complementary 5-block calibration/evaluation halves, learn a single shell-calibrated threshold on one half, and apply it unchanged to the other half at nominal $\alpha = 10^{-3}$. Table 14 shows the result. Under that genuinely fixed threshold transfer, PD and Kasai collapse to zero held-out vessel-core recall, whereas the whitened matched-subspace specialist retains a small but nonzero held-out core TPR of 0.002 [0.000, 0.004] at a realized shell FPR of 0.001 [0.000, 0.002]. The effect remains modest, but it means the in-vivo structural claim does not depend entirely on per-window threshold readjustment.

These real-IQ experiments are best interpreted as bounded structural and robustness audits rather than as definitive labeled *in vivo* task benchmarks. They should also not be read as a replacement for upstream motion correction or registration. Motion correction remains central in ULM and Doppler pipelines because residual motion and clutter leakage can dominate downstream

Window	self-correlation base→STAP	flow hit at bg tail 10^{-3} base→STAP	clip fraction STAP>base (correlation)	clip fraction STAP>base (hit rate)
0:128	$0.188 \to 0.454$	$0.000 \to 0.041$	1.0	1.0
64:192	$0.186 \to 0.399$	$0.000 \to 0.040$	0.9	1.0
122:250	$0.182 \to 0.364$	$0.000 \to 0.041$	0.8	1.0

Table 15: Compact Shin rat-brain open real-IQ corroboration under controlled residual-motion stress (10 clips; 2 px amplitude; rigid registration enabled; KA disabled). "Self-correlation" is the Pearson correlation between each method's score map and its own no-motion reference after alignment and a boundary crop. "Flow-proxy hit rate" is evaluated at a matched background-proxy tail rate of 10^{-3} under a fixed, method-independent flow-proxy mask. The purpose of this table is not to claim task ROC on Shin, but to show that the same nuisance-robustness pattern appears on a second open real-IQ dataset.

vascular estimates [34, 36, 37]. The narrower claim here is that a local nuisance-aware detector head can make the exported score map less brittle to residual motion mismatch after ordinary residualization. That is still practically useful, because many acquisitions fail by producing unstable hot spots and threshold drift rather than by complete loss of flow signal.

Second open real-IQ corroboration: Shin rat-brain sequence. To avoid making the real-IQ story depend on a single open dataset, we also summarize the strongest existing Shin rat-brain motion-stress result from the supplement [31, 32]. Shin does not provide independent flow/background labels, so we do not treat it as a detection benchmark. Instead, under a prespecified low-frequency operating setting and a controlled residual-motion stress test, we ask whether the same qualitative robustness pattern seen on ULM recurs on a second open real-IQ source. It does: across all three audited windows, the STAP-derived score map is more stable than the MC–SVD baseline map under the same 2 px residual-motion stress, and the matched background-tail operating point yields a nonzero flow-proxy hit rate only for STAP.

5.6 Open real-IQ score audit on a common residual

Taken together, the real-data story in this paper is therefore convergent but bounded rather than broad and decisive: the open Twinkling database provides structurally labeled phantom evidence, ULM 7883227 provides open real-IQ motion-robustness evidence, a same-residual score audit, and a bounded localization-derived structural-label result on real in-vivo brain IQ, and the Shin rat-brain sequence provides a second open real-IQ corroboration under controlled residual-motion stress. A different translational question is whether STAP provides a better flow score than PD or Kasai when all three are applied to the same exported real-IQ residual data. We therefore summarize a compact same-residual comparison on the open real-IQ bundles audited here.

In separate retrospective whole-brain label-free sweeps, nuisance-aware alias gating reduced matched-TPR false positives by 18.6% (95% CI 16.8–20.2) while retaining 81.0% of tiles, with a held-out scan check showing a 14.5% reduction (95% CI 7.9–21.4). Those analyses remain label-free and retrospective rather than definitive validation, but they support the same practical interpretation as the open real-IQ evaluations above: nuisance-aware scoring and gating can improve map hygiene without requiring a new acquisition pipeline [9, 43].

Dataset	Residual family	ΔAUC vs PD	ΔAUC vs Kasai	Note
Shin rat-brain IQ sequence	raw	−0.012	+0.004	near-neutral vs Kasai
Shin rat-brain IQ sequence	MC–SVD	−0.134	−0.117	mixed / negative under the audited Shin setting
ULM 7883227 brainlike	MC–SVD-style residual	+0.281	+0.261	positive on all audited bundles
ULM 7883227 elastic	MC–SVD-style residual	+0.282	+0.251	positive on all audited bundles

Table 16: Compact same-residual score audit on the open real-IQ bundles analyzed in this paper. ΔAUC denotes STAP minus the named conventional score on the same residual cube. Coverage is intentionally limited to bundles with active STAP scores: Shin contributes only raw and MC–SVD residual families, while the ULM entries come from the existing MC–SVD-style motion-sweep bundles. The audit supports a strong open real-IQ case for STAP on ULM, but only a mixed one on Shin under the audited prespecified setting.

Supplementary datasets and diagnostics. Additional label-free audits (e.g. whole-brain PD-only retrospectives, Shin contract summaries, and extended motion variants) and diagnostic panels (e.g. conditional-execution leakage ablations, PRF sensitivity, and KA hygiene panels) are provided in Supplementary Material.

6 Discussion

What the detector family adds

The main message of this study is that post-beamformed fUS benefits from a detector-head view of the problem, not only from better residualization. Ordinary clutter suppression can leave a small number of structured false detections that dominate the right tail of the score distribution and disproportionately shape map trustworthiness. The strongest general result in this paper is therefore the simple one: localized band-limited matched-subspace scoring is a stronger fixed default than conventional power-Doppler-style scoring on the held-out structural benchmark. The labeled brain simulations and structurally labeled phantom then explain *why*: they show that the score construction itself matters in the strict tail, and that nuisance-aware local scoring can preserve measurable sensitivity where several conventional score heads collapse.

Those same ablations also define the current boundary of the method family. On the labeled brain simulations, most of the gain comes from band-limited matched-subspace scoring rather than from whitening alone. On the held-out structural benchmark, the unwhitened head remains the strongest fixed nuisance-control choice. The more adaptive modes are still useful, but only in a bounded way: the adaptive mode preserves the default behavior in cleaner structural settings while recovering part of the whitening benefit in clutter-heavy conditions, and the fully whitened mode remains most compelling in the across-view phantom and the held-out functional audit. The practical result is therefore a detector family with a clear fixed default and two more specialized extensions, not a claim that one covariance-whitened score is optimal in every regime.

One deployment point is worth stating explicitly. The strict-tail operating points in the labeled experiments are calibrated using known negative sets, but a scanner does not have frame-by-frame labeled negatives at runtime. In this paper, labeled-tail calibration is therefore an *evaluation protocol*, not the intended clinical operating rule. The more deployment-relevant evidence is now stronger than a purely supplementary diagnostic: the main text reports a separate-bank threshold transfer on the held-out SIMUS/PyMUST benchmark and a complementary block-split fixed-calibration transfer on the PALA-backed ULM audit, both without per-window retuning on the evaluation data. Those checks do not solve cross-session or cross-scanner deployment thresholding outright, but they do show that the central nuisance-control claim survives a genuinely fixed-threshold test on both the main translational benchmark and one bounded real-IQ endpoint. The remaining external step is prospective validation of a practical thresholding workflow on raw *in vivo* IQ without retrospective labels.

Real-data scope and clinical interpretation

The optional shrink-only regularizer is deliberately conservative. It can only shrink scores, it preserves an explicit do-not-penalize set, and it is designed to remain inert when the side information is weak or ambiguous. The calculi audit is therefore meant to show bounded behavior rather than dramatic uplift: the regularizer modestly compresses the extreme tail while leaving protected locations unchanged. That is the right role for this layer in a medical signal-processing pipeline, where bounded failure modes matter more than aggressive apparent gains. It is also why we do not make it part of the main headline claim.

A broader practical limitation is data availability. Most public *in vivo* fUS releases do not preserve the raw high-PRF complex IQ and metadata needed for independent detector evaluation

after beamforming. For that reason, the strongest quantitative efficacy evidence in this paper still comes from labeled simulations and structurally labeled phantom data, while the open real-IQ sections are framed more narrowly as robustness, same-residual, and map-stability audits. The new PALA-backed ULM structural audit partially closes that gap: on a bounded vessel-core versus perivascular-shell endpoint derived from a published localization rendering, the fully whitened matched-subspace specialist is measurably stronger than PD and Kasai, and the prespecified 64-frame no-registration implementation remains below the acquisition-time budget on an RTX 4080 SUPER. The effect size is modest, and it should be read that way. What matters is that the direction is consistent on a real in-vivo structural endpoint with labels derived from a fundamentally different physical process than the Doppler detector itself. That is not the same as independent multimodal validation, and it is not a claim about functional activation or hemodynamic-response recovery, but it is materially stronger than a purely label-free map-stability claim. Even with that remaining limitation, the clinical interpretation is already useful: this is a step toward more trustworthy fUS maps in realistic nuisance and mild-motion conditions, because it reduces artifactual hot spots and nuisance leakage without requiring a new acquisition pipeline. The intended gain is not merely better vessel conspicuity, but more stable decision-relevant maps at matched parenchymal sensitivity when rare false positives would otherwise dominate interpretation.

Detector policy and next validation step

The current evidence supports a simple three-part policy, summarized in Table 17. The fixed band-limited matched-subspace detector is the main result and default for held-out structural use. The adaptive mode is the practical extension when one wants additional clutter suppression without forcing whitening everywhere. The fully whitened variant remains a bounded specialist mode for the regimes in which full covariance adaptation already earns its complexity, especially the across-view phantom, the held-out functional audit, and the bounded PALA-backed in-vivo structural endpoint. The supplementary stress frontier is the clearest reason this hierarchy is needed at all: under the cardiac-reference and shortest-ensemble rows, the fixed matched-subspace default is already best, whereas the harsher pulsation rows and the intermediate short-ensemble row are won by the fully whitened specialist. The PALA-backed ULM result shows the same principle on real data for a different reason: vessel-core concentration relative to the immediately adjacent shell favors the fully whitened matched-subspace specialist, whereas nuisance-control on the SIMUS benchmark favors the simpler fixed head.

This is a more useful result than a generic statement that "whitening might help." It means the paper supports a concrete default method, a practical adaptive extension, and a bounded specialist mode. The main missing external step is now labeled *in vivo* validation on raw high-PRF IQ with independent anatomical or task anchors.

7 Conclusion and future work

We presented a localized matched-subspace detector-head framework for post-beamformed fUS and ultrafast Doppler data, centered on a band-limited matched-subspace fixed head and extended by an adaptive whitening switch plus a bounded fully whitened specialist mode. Across labeled brain stress tests and a structurally labeled phantom, this framework improved strict-tail detection behavior relative to several conventional score heads, while the prespecified held-out SIMUS/PyMUST

Evidence regime	Preferred head	Quantitative anchor	Practical interpretation
Prespecified held-out structural benchmark (mobile and intra-operative parenchymal)	Fixed band-limited matched-subspace head ($R = I$)	Best whitening profile still trails the default: Δselection=-0.173 (mobile) and -0.478 (intra-operative parenchyma)	Use the simpler head as the default translational choice.
Prespecified adaptive variant across labeled and open-data audits	Guard-triggered adaptive mode	Open-skull / structured-clutter TPR@10^{-4} stay at 0.869 / 0.910; mean promoted fraction = 0.111 (Shin) and 0.926 (across-view Gammex)	Preferred adaptive variant when one adaptive score is desired without abandoning the fixed-head behavior in cleaner regimes.
Structurally labeled Gammex phantom (across-view)	Fully whitened covariance-adaptive variant	Across-view TPR@10^{-3}: $0.065 \rightarrow 0.224$	Use the fully whitened variant when maximizing whitening gain matters more than preserving default-like behavior.
Held-out same-residual functional audit	Fully whitened covariance-adaptive variant	Positive in 5/6 mobile families (direct STAP readout) and 4/6 intra-operative families (PD-after-STAP)	Keep the fully whitened variant for function-priority analyses even though it is not the structural default.

Table 17: Detector-family hierarchy implied by the current evidence. The simpler band-limited matched-subspace detector is the fixed default because it remains strongest on the prespecified held-out structural benchmark across all tested fixed profiles. The prespecified guard-triggered adaptive variant preserves that fixed-head behavior in cleaner regimes while selectively turning on whitening in clutter-heavier ones. The fully whitened STAP-inspired variant is retained for settings where full whitening still earns its complexity.

benchmark showed that the simpler band-limited matched-subspace detector is the strongest fixed structural default. A held-out SIMUS stress confirmation then showed that under cardiac-like pulsation and short-ensemble pressure, the adaptive and fully whitened extensions add value on the stressed rows where nuisance pressure is highest. A separate fixed-calibration transfer audit on that same benchmark then showed that thresholds learned on separate calibration seeds still preserve the fixed head's nuisance-control advantage on held-out evaluation seeds without per-window retuning. On real in-vivo rat-brain IQ, a bounded structural-label audit derived from the published PALA localization rendering showed a modest but consistent advantage for the fully whitened matched-subspace specialist over PD and Kasai on a vessel-core versus perivascular-shell endpoint; the same prespecified 64-frame no-registration configuration also remained below the acquisition-time budget on an RTX 4080 SUPER, and a complementary 5-block fixed-calibration transfer retained a small but nonzero held-out core-detection advantage at near-target shell FPR. This in-vivo claim is about structural vascular fidelity rather than functional activation or hemodynamic-response recovery. The real-IQ evidence therefore now includes a quantitative in-vivo structural result, but it remains bounded rather than definitive independent-task validation.

The practical implication is therefore clear. Nuisance-aware matched-subspace scoring is the main deployed consequence of this framework; the adaptive mode is the clearest compromise when

additional clutter suppression is needed, and the fully whitened mode is retained only for the settings in which it already earns its complexity. The near-term clinical implication is pragmatic: a nuisance-robust local detector may help move fUS and ultrafast Doppler outputs from "informative under careful conditions" toward "reliably interpretable under messier real conditions," especially when rare false positives dominate downstream decisions. The main next external test is prospective labeled *in vivo* validation on raw high-PRF IQ with independent anatomical or task anchors and a practical non-retrospective thresholding workflow.

Data and code availability

All datasets used in this work are either publicly available (as cited) or derived deterministically from those sources. All code used to generate the reported tables and figures is versioned; the exact commit hash and release information are recorded in the reproducibility manifest (Supplementary Material). A minimal CUDA-enabled software environment and container recipe are provided in the repository. End-to-end regeneration commands for the reported figures and tables are collected in the repository and summarized in the reproducibility appendix.

Code repository. The reference implementation and reproduction scripts are available at `https://github.com/ArthurShune/stap-for-fus`. The canonical commit hash and release information are recorded in the reproducibility manifest (Supplementary Material).

References

[1] Imbault, M., et al. Intraoperative functional ultrasound imaging of human brain activity. *Scientific Reports* 7, 7304 (2017).

[2] Macé, E., Montaldo, G., Pernot, M., Cohen, I., and Tanter, M. Functional ultrasound imaging of the brain. *NeuroImage* 54(4), 2720–2731 (2011).

[3] Urban, A., et al. In vivo whole brain microvascular imaging in mice using ultrafast ultrasound. *EBioMedicine* 76, 103861 (2022).

[4] Alexandrov, A. V., et al. Velocity measurements in the middle cerebral arteries of healthy adults by transcranial Doppler. *Journal of Neuroimaging* 1, 166–172 (1991).

[5] Alexandrov, A. V., et al. Optimal values of flow velocity on transcranial Doppler in patients with intracranial arterial stenosis. *Stroke* 33, 1757–1761 (2002).

[6] Macé, E., et al. Functional ultrasound imaging of the brain: theory and basic principles. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 60(3), 492–506 (2013).

[7] Rabut, C., et al. Functional ultrasound imaging of human brain activity through the skull. *Science Translational Medicine* 12(572), eaav4392 (2020).

[8] Rabut, C., Norman, S. L., Griggs, W. S., Russin, J. J., Jann, K., et al. Functional ultrasound imaging of human brain activity through an acoustically transparent cranial window. *Science Translational Medicine* 16, eadj3143 (2024).

[9] Deffieux, T., Demené, C., and Tanter, M. Functional ultrasound imaging: A new imaging modality for neuroscience. *Neuroscience* 474, 110–121 (2021).

[10] Demeulenaere, O., Bertolo, A., Pezet, S., Osmanski, B.-F., Tanter, M., and Deffieux, T. In vivo whole brain microvascular imaging in mice using transcranial 3D ultrasound localization microscopy. *EBioMedicine* 79, 103995 (2022).

[11] Demené, C., et al. Spatiotemporal clutter filtering of ultrafast ultrasound data highly increases Doppler and fUltrasound sensitivity. *IEEE Transactions on Medical Imaging* (2015). DOI: `https://doi.org/10.1109/TMI.2015.2428634`.

[12] Baranger, J., et al. Adaptive spatiotemporal SVD clutter filtering for ultrafast Doppler imaging using similarity of spatial singular vectors. *IEEE Transactions on Medical Imaging* (2018). DOI: `https://doi.org/10.1109/TMI.2018.2789499`.

[13] Yu, A. C. H., and Løvstakken, L. Eigen-based clutter filter design for ultrasound color flow imaging: a review. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* (2010). DOI: `https://doi.org/10.1109/TUFFC.2010.1521`.

[14] Song, P., et al. Ultrasound small vessel imaging with block-wise adaptive local clutter filtering. *IEEE Transactions on Medical Imaging* (2017). DOI: `https://doi.org/10.1109/TMI.2016.2605819`.

[15] Kasai, C., Namekawa, K., Koyano, A., and Omoto, R. Real-time two-dimensional blood flow imaging using an autocorrelation technique. *IEEE Transactions on Sonics and Ultrasonics* 32(3), 458–464 (1985). DOI: `https://doi.org/10.1109/T-SU.1985.31615`.

[16] Chee, A. J. Y., Yiu, B. Y. S., and Yu, A. C. H. A GPU-parallelized eigen-based clutter filter framework for ultrasound color flow imaging. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* (2017). DOI: `https://doi.org/10.1109/TUFFC.2016.2606598`.

[17] Lok, U.-W., Tang, J., Kilic, K., Szabo, T. L., and Boas, D. A. Real time SVD-based clutter filtering using randomized SVD and spatial downsampling for microvessel imaging. *Ultrasonics* 107, 106163 (2020). DOI: `https://doi.org/10.1016/j.ultras.2020.106163`.

[18] Demene, C., et al. Improved color Doppler for cerebral blood flow axial velocity estimation (iCD-fUS). *IEEE Transactions on Medical Imaging* 40(5), 1498–1510 (2021).

[19] Tang, J., Kilic, K., Szabo, T. L., and Boas, D. A. Improved color Doppler for cerebral blood flow axial velocity imaging. *IEEE Transactions on Medical Imaging* 40(2), 758–764 (2021).

[20] Demené, C., et al. Functional ultrasound imaging of brain activity in human newborns. *Science Translational Medicine* 9(411), eaah6756 (2017). DOI: `https://doi.org/10.1126/scitranslmed.aah6756`.

[21] Baranger, J., et al. Bedside functional monitoring of the dynamic brain connectivity in human neonates. *Nature Communications* 12, 1080 (2021). DOI: `https://doi.org/10.1038/s41467-021-21387-x`.

[22] Soloukey, S., et al. Mobile human brain imaging using functional ultrasound. *Science Advances* 11(25), eadu9133 (2025). DOI: `https://doi.org/10.1126/sciadv.adu9133`.

[23] Aaslid, R., Markwalder, T.-M., and Nornes, H. Noninvasive transcranial Doppler ultrasound recording of flow velocity in basal cerebral arteries. *Journal of Neurosurgery* 57(6), 769–774 (1982).

[24] Gao, S., Lam, W. W.-K., Chan, Y.-H., Liu, J., Wong, K.-P., Leung, T. W.-S., et al. Optimal values of flow velocity on transcranial Doppler in grading middle cerebral artery stenosis in comparison with magnetic resonance angiography. *Journal of Neuroimaging* 12(3), 213–218 (2002).

[25] Estrada, H., Gottschalk, S., Reiss, M., Neuschmelting, V., Razansky, D., et al. Observation of guided acoustic waves in a human skull. *Ultrasound in Medicine & Biology* 44(7), 1438–1447 (2018).

[26] Candès, E. J., Li, X., Ma, Y., and Wright, J. Robust principal component analysis? *Journal of the ACM* 58(3), 11 (2011).

[27] Solomon, O., et al. Deep unfolded robust PCA with application to clutter suppression in ultrasound. *IEEE Transactions on Medical Imaging* 39(4), 1051–1063 (2020). DOI: `https://doi.org/10.1109/TMI.2019.2941271`.

[28] Lin, Z., Chen, M., and Ma, Y. The augmented Lagrange multiplier method for exact recovery of corrupted low-rank matrices. *arXiv preprint* arXiv:1009.5055 (2010).

[29] Halko, N., Martinsson, P. G., and Tropp, J. A. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review* 53(2), 217–288 (2011).

[30] Soloukey, S., Vincent, A. J. P. E., Smits, M., Slump, C. H., Dirven, C. M. F., et al. Functional ultrasound (fUS) during awake brain surgery: the clinical potential for intra-operative functional and vascular brain mapping. *Frontiers in Neuroscience* 13, 1384 (2019). DOI: `https://doi.org/10.3389/fnins.2019.01384`.

[31] Shin, Y., et al. Context-Aware Deep Learning Enables High-Efficacy Localization of High Concentration Microbubbles for Super-Resolution Ultrasound Localization Microscopy. Zenodo dataset, DOI 10.5281/zenodo.10711806 (2024).

[32] Shin, Y., et al. Context-Aware Deep Learning Enables High-Efficacy Localization of High Concentration Microbubbles for Super-Resolution Ultrasound Localization Microscopy. *Nature Communications* 15, 3299 (2024). DOI: `https://doi.org/10.1038/s41467-024-47154-2`.

[33] Chavignon, A., Heiles, B., Hingot, V., Lopez, P., Teston, E., and Couture, O. In vivo rat brain for Ultrasound Localization Microscopy: raw and beamformed data. Zenodo dataset, DOI 10.5281/zenodo.7883227 (2023).

[34] Heiles, B., Chavignon, A., Hingot, V., Lopez, P., Teston, E., and Couture, O. Performance benchmarking of microbubble-localization algorithms for ultrasound localization microscopy. *Nature Biomedical Engineering* (2021). DOI: https://doi.org/10.1038/s41551-021-00824-8.

[35] Errico, C., et al. Ultrafast ultrasound localization microscopy for deep super-resolution vascular imaging. *Nature* 527(7579), 499–502 (2015). DOI: https://doi.org/10.1038/nature16066.

[36] Hingot, V., et al. Subwavelength motion-correction for ultrafast ultrasound localization microscopy. *Ultrasonics* 77, 17–21 (2017). DOI: https://doi.org/10.1016/j.ultras.2017.01.008.

[37] Insana, M. F., Dai, B., Babaei, S., and Abbey, C. K. Combining spatial registration with clutter filtering for power-Doppler imaging in peripheral perfusion applications. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 69(12), 3243–3254 (2022). DOI: https://doi.org/10.1109/TUFFC.2022.3211469.

[38] Leonov, D. V., Reshetnikov, R. V., Kulberg, N. S., Nasibullina, A. A., and Gromov, A. I. Doppler Twinkling Artifact Observations: an Open-Access Database of Raw Ultrasonic Signals. *Digital Diagnostics* (2021). DOI: https://doi.org/10.17816/DD76511.

[39] Ozgun, K., and Byram, B. C. Multidimensional clutter filtering of aperture domain data for improved blood flow sensitivity. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 68(8), 2645–2656 (2021). DOI: https://doi.org/10.1109/TUFFC.2021.3073292.

[40] Huang, Z., et al. Adaptive higher-order singular value decomposition clutter filter for ultrafast Doppler imaging of coronary flow under non-negligible tissue motion. *Ultrasonics* (2024). DOI: https://doi.org/10.1016/j.ultras.2024.107307.

[41] Garcia, D. SIMUS: an open-source simulator for medical ultrasound imaging. Part I: theory and examples. *Computer Methods and Programs in Biomedicine* 218, 106726 (2022). DOI: https://doi.org/10.1016/j.cmpb.2022.106726.

[42] Cigier, A., Varray, F., and Garcia, D. SIMUS: an open-source simulator for medical ultrasound imaging. Part II: comparison with four simulators. *Computer Methods and Programs in Biomedicine* 220, 106774 (2022). DOI: https://doi.org/10.1016/j.cmpb.2022.106774.

[43] Urban, A., et al. Real-time imaging of brain activity in freely moving rodents using functional ultrasound. *Nature Methods* 12(9), 873–878 (2015). DOI: https://doi.org/10.1038/nmeth.3482.

[44] Demené, C., et al. 4D microvascular imaging based on ultrafast Doppler tomography. *NeuroImage* 127, 472–483 (2016). DOI: https://doi.org/10.1016/j.neuroimage.2015.11.014.

[45] Slepian, D. Prolate spheroidal wave functions, Fourier analysis, and uncertainty—V: The discrete case. *Bell System Technical Journal* 57(5), 1371–1430 (1978). DOI: `https://doi.org/10.1002/j.1538-7305.1978.tb02104.x`.

[46] Thomson, D. J. Spectrum estimation and harmonic analysis. *Proceedings of the IEEE* 70(9), 1055–1096 (1982). DOI: `https://doi.org/10.1109/PROC.1982.12433`.

[47] Percival, D. B., and Walden, A. T. *Spectral Analysis for Physical Applications: Multitaper and Conventional Univariate Techniques*. Cambridge University Press (1993).

[48] Guerci, J. R. *Space-Time Adaptive Processing for Radar*. Artech House (2003).

[49] Wicks, M. C., Rangaswamy, M., Adve, R., and Hale, T. B. Space-time adaptive processing: a knowledge-based perspective for airborne radar. *IEEE Signal Processing Magazine* 23(1), 51–65 (2006). DOI: `https://doi.org/10.1109/MSP.2006.1593337`.

[50] Melvin, W. L., and Guerci, J. R. Knowledge-aided signal processing: a new paradigm for radar and other advanced sensors. *IEEE Transactions on Aerospace and Electronic Systems* 42(3), 983–996 (2006). DOI: `https://doi.org/10.1109/TAES.2006.248215`.

[51] Gurram, P. R., and Goodman, N. A. Spectral-domain covariance estimation with a priori knowledge. *IEEE Transactions on Aerospace and Electronic Systems* 42, 1010–1020 (2006). DOI: `https://doi.org/10.1109/TAES.2006.248209`.

[52] Van Veen, B. D., and Buckley, K. M. Beamforming: a versatile approach to spatial filtering. *IEEE ASSP Magazine* 5(2), 4–24 (1988). DOI: `https://doi.org/10.1109/53.665`.

[53] Kuglin, C. D., and Hines, D. C. The phase correlation image alignment method. In *Proceedings of the IEEE International Conference on Cybernetics and Society* (1975).

[54] Stone, H. S., Orchard, M. T., Chang, E.-C., and Martucci, S. A. A fast direct Fourier-based algorithm for subpixel registration of images. *IEEE Transactions on Geoscience and Remote Sensing* 39(10), 2235–2243 (2001). DOI: `https://doi.org/10.1109/36.957286`.

[55] Foroosh, H., Zerubia, J. B., and Berthod, M. Extension of phase correlation to subpixel registration. *IEEE Transactions on Image Processing* 11(3), 188–200 (2002). DOI: `https://doi.org/10.1109/83.988953`.

[56] Capon, J. High-resolution frequency-wavenumber spectrum analysis. *Proceedings of the IEEE* 57(8), 1408–1418 (1969).

[57] Frost, O. L. An algorithm for linearly constrained adaptive array processing. *Proceedings of the IEEE* 60(8), 926–935 (1972).

[58] Kay, S. M. *Fundamentals of Statistical Signal Processing, Volume II: Detection Theory*. Prentice Hall (1998).

[59] Scharf, L. L., and Friedlander, B. Matched subspace detectors. *IEEE Transactions on Signal Processing* 42(8), 2146–2157 (1994).

[60] Ward, J. Space-time adaptive processing for airborne radar. MIT Lincoln Laboratory, Tech. Rep. 1015 (1994).